Niayesh Afshordi

Astrophysics, cosmology, and gravity

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Comprehensive public academic CV — updated 10 November 2026

Profile

Niayesh Afshordi is an astrophysicist working at the interface of cosmology, gravitation, quantum theory, and observation. His research connects fundamental questions to gravitational-wave data, black holes, the cosmic microwave background, large-scale structure, compact objects, and other astrophysical probes. Current themes include quantum black holes and quadratic gravity, strong-gravity tests, early-universe cosmology, dark matter and dark energy, missing baryons, and modified gravity. His interdisciplinary work also includes medical-physics applications, radiotherapy, and epidemiological and public-health modelling.

Research profiles

- arXiv author record
- INSPIRE author record
- ORCID 0000-0002-9940-7040
- Google Scholar profile

Scope

This long public CV retains the complete professional record, including appointments, funding, teaching, supervision, service, presentations, and interviews, while excluding protected personal information. The publication list is refreshed independently from arXiv, INSPIRE, and ORCID. The separate website talks collection holds source decks and slide exports; raw filenames, duplicate export formats, and other speakers' files are not CV entries.

Professional record

Generated from the Canadian Common CV submitted on 15 August 2025. The protected cover page and personal contact, demographic, citizenship, and identifier fields are excluded. Administrative form metadata is suppressed for readability; the substantive academic record is retained.

Privacy screen: 3 protected source line(s) removed in addition to the protected cover page.

Language Skills

English and Persian: read, write, speak, understand, and peer review.

Degrees

2000/9 - 2004/11	Doctorate, PhD, Astrophysics, Princeton University Thesis Title: The Other 99 Percent Supervisors: Spergel, David, 2000/9 - 2004/11 Research Disciplines: Astronomy and Astrophysics
1996/9 - 1999/5	Bachelor's Honours, BSc, Physics, Sharif University of Technology

Recognitions

2020/1	The Buchalter Cosmology Prize 2019 Dr. Ari Buchalter Prize / Award The $10,000 First Prize was awarded to Dr. Jahed Abedi of the Max-Planck-Institut fur Gravitationsphysik (Albert-Einstein-Institut) and Leibniz Universitat Hannover and Dr. Niayesh Afshordi of the University of Waterloo and the Perimeter Institute for Theoretical Physics, for their work entitled "Echoes from the Abyss: A highly spinning black hole remnant for the binary neutron star merger GW170817", published in the Journal of Cosmology and Astroparticle Physics and recognized by the judging panel as “a bold and innovative step towards understanding quantum gravitational phenomena, using observational data to demonstrate gravitational wave echoes from a binary neutron star merger, which could indicate that black hole interiors may be more complex than the simple regions predicted by General Relativity.”
2016/1	The Buchalter Cosmology Prize 2015 Dr. Ari Buchalter Prize / Award The $2,500 Third Prize was awarded to Dr. Niayesh Afshordi of the University of Waterloo and the Perimeter Institute for Theoretical Physics and Dr. Elliot Nelson of Perimeter Institute, for their work entitled “Cosmological Non-Constant Problem: Cosmological bounds on TeV-scale physics and beyond”, published in the Journal of Physical Review D and recognized by the judging panel as "an intriguing proposition that the Planck scale of quantum gravity may soon be accessible by particle accelerators, based on the expected white noise fluctuations in the cosmological constant, which might not be constant at all but vary widely in space.”
2011/10	Professor M.K.Vainu Bappu Gold Medal Astronomical Society of India Prize / Award The Vainu Bappu Memorial Gold Medal honours outstanding astronomers worldwide under the age of 35. I was one of the two recipients of this honour in 2011.
2011/9	Early Researcher Award - 150,000 (Canadian dollar) Government of Ontario Prize / Award The ERA program helps promising, recently-appointed Ontario researchers build their research teams of undergraduates, graduate students, post-doctoral fellows, research assistants, associates and technicians. The goal of the program is to improve Ontario’s ability to attract and retain the best and brightest research talent.
2010/9 - 2013/9	Discovery Accelerator Supplement - 120,000 Natural Sciences and Engineering Research Council of Canada (NSERC) Prize / Award The Discovery Accelerator Supplement award provides substantial and timely resources to a small group of researchers whose research proposals suggest and explore highrisk, novel or potentially transformative concepts and lines of inquiry, and are likely to have impact by contributing to groundbreaking advances in the area. I was one of the 8 researchers in Physics who received this award in 2010.

User Profile

Researcher Status: Researcher Research Career Start Date: 1998/01/01 Engaged in Clinical Research?: No

Key Theory / Methodology General Relativity, Quantum Mechanics, Astrophysical and Cosmological Observations, Numerical Simulations, Large Data, Statistical Methods, Theoretical Modeling

Research Interests I lead a comprehensive and ambitious research program, which spans a wide area, from Astrophysics/Cosmology to Fundamental Physics, and includes a diverse spectrum of observational, numerical, and analytic elements. The program aims at linking deep mysteries in fundamental physics to novel observational results and techniques, which can be used to shed light on the structure and laws of our Universe. The areas that my group and I currently focus on are: 1. Phenomenology of Dark Energy and cosmological modifications of General Relativity 2. Distribution of Dark Matter on small scales and novel windows for Dark Matter detection 3. Observational consequences of Aether theories of gravity, and their connection to Quantum Gravity and Cosmological Constant Problem 4. Early Universe, and its links to foundations of Gravity and Quantum Mechanics 5. Missing baryons, and observational probes of intergalactic plasma 6. Physics of Neutron Stars, Black Holes and Accretion Disks.

Research Experience Summary I am an Astronomer, and a Physicist; in short I am an Astrophysicist. Currently, I am a Professor in the Astrophysics and Gravitation group at the Department of Physics and Astronomy, University of Waterloo. I am also Research Associate Faculty in the Cosmology and Gravitation group at the Perimeter Institute for Theoretical Physics (PI), the premier Canadian institute for research in theoretical Physics. I finished my PhD at

Princeton University Observatory in August 2004. I was then a fellow at the Institute for Theory and Computation (ITC) at the Harvard-Smithsonian Center for Astrophysics (2004-2007), and a Distinguished Research Fellow at PI (2008-2009). To date, I have published more than 120 peer-reviewed articles, which have been cited more than 7600 times. I have also (co-)supervised 82 students (40 of them at the doctoral level), as well as 28 postdoctoral fellows. Moreover, I have co-organized 31 international scientific meetings/seminar series, and raised over 1.5 million dollars in external funding.

Research Specialization Keywords Astrophysics, Cosmology, Gravitational Waves, Physics of Gravity, Cosmic Microwave Background, Dark Matter, Dark Energy, Black Hole Physics and Astrophysics, Missing Baryons, Lorentz Violation, Quantum Gravity, COVID-19 epidemeology, Flash Radiotherapy

Disciplines Trained In Astronomy and Astrophysics, Physics

Research Disciplines Astronomy and Astrophysics, Physics, Epidemiology and Biostatistics, Radiology

Areas of Research Galactic and Cosmic, Universe Structure, Gravitation, Quantum Phenomena, Elementary Particles

Fields of Application Foundations and Knowledge Acquisition, Biomedical Aspects of Human Health, Public Health

Employment

2022/9	Full Professor of Physics and Astronomy Physics and Astronomy, University of Waterloo Full-time, Professor Professor in the Astrophysics and Gravitation group. Founding Member of the Waterloo Centre for Astrophysics (2018)
2009/9	Research Associate Faculty Cosmology and Gravitation, Waterloo, Perimeter Institute for Theoretical Physics A Renewable appointment as a Perimeter Institute for Theoretical Physics Faculty Member.
2015/9 - 2022/7	Associate Professor Physics and Astronomy, Waterloo Centre for Astrophysics, University of Waterloo Full-time, Associate Professor Associate Professor in the Astrophysics and Gravitation group. Founding Member of the Waterloo Centre for Astrophysics (2018)
2009/9 - 2015/9	Assistant Professor Physics and Astronomy, Science, University of Waterloo Full-time, Assistant Professor Assistant Professor in the Astrophysics and Gravitation group
2012/9 - 2013/8	Visiting Professor Physics, North, SUNY at Buffalo Part-time, Visiting Professorship I visited this department on a regular basis and engaged and collaborated with the faculty in Cosmology.
2007/9 - 2009/8	Distinguished Research Fellow Cosmology and Gravitation, Waterloo, Perimeter Institute for Theoretical Physics
2004/9 - 2007/8	Postdoctoral Fellow Astrophysics, Institute for Theory and Computation, Harvard-Smithsonian Center for Astrophysics
2000/9 - 2004/8	Graduate Student Instructor and Researcher Astrophysical Sciences, Arts and Sciences, Princeton University
1999/9 - 2000/8	Graduate Student Teaching Assistant Physics, Arts and Sciences, Brown University
1998/9 - 1999/8	Research Associate Physics, Tehran, Institute for Theoretical Studies in Physics and Mathematics (IPM)
1996/1 - 1998/8	Instructor Iranian National Physics Olympiad Camp, International Physics Olympiad

Affiliations

The primary affiliation is denoted by (*)

(*) 2015/9	Associate Professor, University of Waterloo

2009/9 Research Associate Faculty, Perimeter Institute for Theoretical Physics

Leaves of Absence and Impact on Research

2019/9 - 2020/8 Sabbatical, University of Waterloo I was on sabbatical at Perimeter Institute, and did not have any research/teaching duties at U-Waterloo.

Research Funding History

Current

2025/4 - 2030/3 **NSERC Discovery Grant** — *Quantum Black Holes, Vacuum, and the Big Bang: Unraveling Cosmic Mysteries with Multi-Messenger Astronomy.* Principal Investigator. **C$250,000**.

Completed

2022/9 - 2025/8 **University of Waterloo Interdisciplinary Trailblazer Grant** — *Quantum Spacetime in the Cosmos: From Conception to Reality.* Principal Investigator; Co-investigator: Doreen Fraser; Collaborator: Steven Weinstein. **C$80,000**.

2021/9 - 2023/8 **Arthur B. McDonald Institute HQP Pooled Resources Competition**. Principal Investigator. **C$40,000**.

2020/4 - 2025/3 **NSERC Discovery Grant** — *Novel Astrophysical Windows into Fundamental Physics: From Big Bang to Black Holes, and the Dark Universe.* Principal Applicant. **C$170,000**.

2015/4 - 2020/3 **NSERC Discovery Grant** — *Novel Astrophysical Windows into Fundamental Physics.* Principal Investigator. **C$210,000**.

2011/4 - 2016/3 **Ontario Early Researcher Award** — *Astrophysical Windows into Fundamental Physics.* Principal Applicant. **C$150,000**.

2010/4 - 2015/3 **NSERC Discovery Grant** — *Astrophysical Windows into Fundamental Physics.* Principal Applicant. **C$280,000**.

2010/4 - 2013/3 **NSERC Discovery Accelerator Supplement** — *Astrophysical Windows into Fundamental Physics.* Principal Applicant. **C$120,000**.

Courses Taught

2026/2 - 2026/3 PHYS 621: Cosmology, Perimeter Scholars International, Perimeter Institute; recorded lectures on 13 and 17 March 2026 (PIRSA:26030037 and PIRSA:26030038).

2025/9 - 2026/4 SCI 238: Introductory Astronomy, University of Waterloo (Fall 2025, Spring 2026, Winter 2026; Winter 2026 co-taught with Mike Hudson)

2013/1 - 2025/4 SCI 238: Introductory Astronomy, University of Waterloo (Winter/Spring 2013, Winter 2015, Winter 2019, Winter/Spring 2021, Spring 2022, Spring 2023, Spring 2025)

2021/9 PHYS 781: Fundamentals of Astrophysics, University of Waterloo (Fall 2021)

2013/9 - 2017/8 PHYS 781: Fundamentals of Astrophysics, University of Waterloo (Fall 2013–Fall 2017)

2018/9 PHYS 475: Astrophysics and Cosmic Structure, University of Waterloo (Fall 2018)

2011/9 PHYS 703: Introduction to Quantum Field Theory, University of Waterloo and Perimeter Institute (Fall 2011)

2010/1 - 2011/4	PHYS 275: Introduction to Astronomy, University of Waterloo (Spring 2010, Spring 2011)
2009/9	Cosmology: Expanding Universe, Perimeter Scholars International (2009)
2008/9	Astrophysics and Cosmology through Problems, University of Waterloo (Fall 2008)

Student/Postdoctoral Supervision

Current group — October 2026

Doctoral	Alice Chen; Ruolin Liu; Ohoud K. Alharbi; Robin Oberfrank; Batia Friedman-Shaw (co-supervised).
Master's	Hassan Subhi (MMath); Harrison (Haoye) Li (MSc).
Postdoctoral	Shokoufe Faraji.
Undergraduate	Aliv Sahoo (research intern); Noni Hajatri; Annika Cheng; Owen Roth.

Individual research projects and completed appointments are recorded below. Yusuke Manita's postdoctoral group appointment ended in 2026; he is an alumnus.

Bachelor's

Current	Hajatri, Noni, University of Waterloo. Undergraduate research on the richness–mass relation of galaxy clusters, working with Alice Chen and Lammim Ahad.
Current	Cheng, Annika, University of Waterloo. Undergraduate research on tidal disruption events, working with Shokoufe Faraji.
Current	Roth, Owen, University of Waterloo. Undergraduate research on modified gravity in binary black hole mergers.
2023/9 - 2025/8	Rasheed, Nabeel, University of Waterloo Principal Supervisor Thesis/Project Title: NSERC USRA project: "Analytic models for gravitational wave form templates" Present Position: PhD Student, Massachusetts Institute of Technology
2026/9	Sahoo, Aliv, University of Waterloo, Undergraduate research intern. Present Position: Undergraduate Student, Indian Institute of Science (IISc)
2023/9 - 2024/7	Yin, Xitai (John), University of Waterloo Principal Supervisor Thesis/Project Title: Black Holes in Quantum Gravity Present Position: MSc Physics – Quantum Technology, University of Waterloo (2025)
2023/9 - 2025/8	Tang, Ruijing, University of Waterloo Principal Supervisor Thesis/Project Title: BSc thesis project: "Exploring Gravitational Effects on accretion disk dynamics in the Teleparallel Born-Infeld gravity" Present Position: Graduate Student, Paris Observatory
2021/9 - 2023/8	Liu, Ruolin, University of Waterloo Principal Supervisor Thesis/Project Title: BSc Thesis Project: "Detecting cosmological scalar fields using orbital networks of quantum sensors" Present Position: PhD Student, University of Waterloo and Perimeter Institute
2022/1 - 2025/4	Subhi, Hassan, University of Waterloo Principal Supervisor Student Degree Expected Date: 2025/4 Thesis/Project Title: Research Project: "Gravitational Wave Echoes in Pulsar Timing Arrays" Present Position: MMath Student, University of Waterloo and Perimeter Institute
2023	Kao, Michelle, University of Waterloo Principal Supervisor Undergraduate research co-op in astrophysics and cosmology. Present Position: MSc student in Aerospace Engineering, Carleton University (from Fall 2026)

2021/2 - 2021/9 Zheng, Jolene, University of Waterloo Principal Supervisor Thesis/Project Title: Covid forecast modeling and online dashboard Present Position: Software Engineer, Bloomberg

2021/2 - 2021/9 Olalekan Koiki, Shafika, University of Waterloo Principal Supervisor Thesis/Project Title: Co-op project: "Covid forecast modeling and online dashboard"

2021/1 - 2021/8 Wisniewski, Kasia, University of Waterloo Principal Supervisor Thesis/Project Title: BSc thesis project: "Searching for Black Hole Echoes from the Failed Supernova AT2017be" Present Position: Satellite Operations Technician, GHGSat

Bachelor's Honours

2024/9 - 2025/5 Shah, Saral, University of Waterloo Principal Supervisor Student Degree Expected Date: 2025/5 Thesis/Project Title: Bachelor's thesis: "Dynamical Friction in a neutrinos sea" Present Position: Undergraduate Student, University of Waterloo

2024/9 - 2025/5 Yanover, Lyvia, University of Waterloo Principal Supervisor Student Degree Expected Date: 2025/5 Thesis/Project Title: Bachelor's thesis: "Bi-Thermal Big Bang" Present Position: MSc Student, University of Waterloo

2024/9 - 2025/5 Murali, Nikhil, University of Waterloo Principal Supervisor Student Degree Expected Date: 2025/5 Thesis/Project Title: Bachelor's Thesis: "Quantum Black Holes: quantum corrections at the tip of a cigar" Present Position: Undergraduate Student, University of Waterloo

2024/9 - 2025/5 Brisebois, Alex Dominic Raul, University of Waterloo Principal Supervisor Student Degree Expected Date: 2025/5 Thesis/Project Title: Bachelor's thesis: "Clustering in the Action-Angle Space" Present Position: PhD Student, University of Toronto

2023/9 - 2024/1 Zhao, Guandi, Fudan University, China Principal Supervisor Thesis/Project Title: BSc visiting student at Perimeter Institute for Theoretical Physics: "Thermal SZ contribution from Low mass galaxies in the unWISE survey" Present Position: MSc student, ETH Zurich, Switzerland

2023/9 - 2024/8 Goel, Lakshay, University of Waterloo Co-Supervisor Thesis/Project Title: Accelerated detector in a superposed spacetime Present Position: Co-founder and Team Lead, MedullaAI

2023/9 - 2024/7 Gifford, Hazel, University of Waterloo Principal Supervisor Thesis/Project Title: Bohmian Rhapsody: Cuscuton and Cosmology Present Position: MSc Student, Brock University

2022/9 - 2023/6 Schiebelbein-Zwack, Aryanna, University of Waterloo Principal Supervisor Thesis/Project Title: BSc Thesis Projects: "Mystery noise in LIGO" Present Position: PhD Student, University of Toronto and CITA

2022/1 - 2024/7 Li, Yu, University of Waterloo Principal Supervisor Thesis/Project Title: NSERC USRA project: "Detecting cosmological scalar fields using orbital networks of quantum sensors" Present Position: PhD Student, California Institute of Technology

2021/5 - 2023/8 Lawrence, Jensen Pierce Thomas, University of Waterloo Principal Supervisor Thesis/Project Title: Research Project: "Universal horizon formation in the incompressible limit of Einstein-Aether theory" Present Position: PhD Student, Department of Earth, Atmospheric and Planetary Sciences, Massachusetts Institute of Technology

2020/9 - 2023/8 Wen, Robin, Department of Applied Math, University of Waterloo Principal Supervisor Thesis/Project Title: BSc Thesis project: "A Cosmic Glitch in gravity" Present Position: PhD Student, California Institute of Technology

2020/9 - 2021/8 Stewart, Meagan, University of Waterloo Principal Supervisor Thesis/Project Title: BSc thesis project: "Inferring Transient Weak Lensing from Quasar Light Curves" Present Position: MSc Student, Simon Fraser University, British Columbia

2020/5 - 2022/5 Shawqi, Shafayat, University of Waterloo Principal Supervisor Thesis/Project Title: BSc Thesis Project: "Sorkin-Johnston Effective Field Theory" Present Position: Graduate Research Assistant in Physics, University of Alberta

2020/1 - 2020/5 Duff, Petra Mary, University of Waterloo Principal Supervisor Thesis/Project Title: NSERC USRA project: "Echoes from the Abyss" Present Position: PhD Student, University of Toronto

2019/9 - 2022/8 Minhas, Palka, University of Waterloo Principal Supervisor Thesis/Project Title: Fast Radio Bursts Present Position: MSc Student, McGill University

2017/1 - 2017/4 Adair, Connor, University of Waterloo Principal Supervisor Thesis/Project Title: NSERC USRA recipient working on Gravitational Wave Echoes Present Position: MSc Student, University of British Columbia

2016/1 - 2016/4 Dykaar, Hannah, University of Waterloo Principal Supervisor Thesis/Project Title: NSERC USRA research project: "Echoes from the Abyss" Present Position: PhD Student, University of Toronto

2016/1 - 2016/4 Elgood, Zachary, University of Waterloo Principal Supervisor Thesis/Project Title: NSERC USRA recipient working on Einstein-Aether Black Holes Present Position: La Caixa Doctoral Scholar, Institute of Theoretical Physics (IFT), Madrid, Spain

2010/9 - 2011/6 Bovard, Luke, University of Waterloo Co-Supervisor Thesis/Project Title: Undergraduate B.Sc. Honours Research Project Present Position: Senior Data Scientist, Plato

2008/9 - 2009/8 Mueller, Matthias, Ulm University Co-Supervisor Thesis/Project Title: Undergraduate Research Project Present Position: Scientist, Forschungszentrum Juelich

Master's non-Thesis

2026/9 - 2028/8 Li, Harrison (Haoye), University of Waterloo, Principal Supervisor, MSc student

2023/9 - 2025/8 Liu, Ruolin, University of Waterloo Principal Supervisor Thesis/Project Title: Quadratic Gravity in Big Bang and Black Holes Present Position: PhD Student, University of Waterloo and Perimeter Institute

2023/3 - present Friedman-Shaw, Batia, Perimeter Institute for Theoretical Physics/ University of Waterloo Co-supervisor Thesis/Project Title: MSc Project: "Doppler Bias: Impact of Peculiar Velocities on Color Selection and the Large Scale Structure of Galaxy Surveys" Present Position: PhD Student

2022/9 - 2023/6 Hrytseniak, Oleksandra, Perimeter Institute for Theoretical Physics/ Co-Supervisor University of Waterloo Thesis/Project Title: Project supervision: "Search for flattened non-gaussianity" Present Position: Resident PhD Student, Perimeter Institute and University of Waterloo

2019/9 - 2020/8 Chua, Wan Zen, University of Waterloo Principal Supervisor Thesis/Project Title: PSI Research Project: "Electromagnetic Albedo of Fuzzballs" Present Position: PhD student, Cornell University

2015/9 - 2016/6 Wang, Qingwen, Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Fuzzball approach to black holes Present Position: XVA Quant, Scotiabank

2013/8 - 2014/6 Roberts, Brodyn, University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: PSI Research Project Present Position: Software Engineer, Index Exchange, Toronto, Canada

2010/9 - 2012/6 Agarwal, Abhineet, University of Waterloo Principal Supervisor Thesis/Project Title: Thermal Tachyacoustic Cosmology Present Position: PhD, Jamia Millia Islamia (2020)

Master's Thesis

2022/9 - 2024/8 Yang, Jiayue, University of Waterloo Co-Supervisor Thesis/Project Title: Entropy of Analogue Black Holes in Tilted Dirac Cone Materials Present Position: PhD Student, University of Waterloo

2021/1 - 2021/12 Golafshani, Hamid (Withdrawn), University of Waterloo Principal Supervisor Thesis/Project Title: Topics in Astrophysics and Cosmology Present Position: Professor of Mechanical Engineering, Conestoga College, Waterloo, ON

2018/9 - 2020/8 Hergott, Samantha, University of Waterloo Principal Supervisor Thesis/Project Title: Unifying Big Bang and Black Holes Present Position: PhD, York University and Perimeter Institute (2025)

2018/9 - 2019/8 — Yang, Tianyi, University of Waterloo Principal Supervisor Thesis/Project Title: Cosmic Filaments, Galactic Kinematics in Gaia Present Position: PhD Student, University of Edinburgh

2018/9 - 2020/8 — Chen, Alice, University of Waterloo Principal Supervisor Thesis/Project Title: Cosmological Structure Formation Present Position: PhD Student, University of Waterloo

2017/9 - 2018/7 — Boruah, Supranta, University of Waterloo Co-Supervisor Thesis/Project Title: Constraining the mass of Milky Way based on clustering in the action space of Gaia DR2 stars Present Position: Center for Particle Cosmology Fellow, University of Pennsylvania

2016/9 - 2017/8 — Kim, Hyungjin (Tony), University of Waterloo Principal Supervisor Thesis/Project Title: The Early Universe: New aspects of Cuscuton Theory, Modified Dispersion Relations and Pulsar Constraints on Vacuum Noise Present Position: Software Engineer, Validus Risk Management

2014/9 - 2016/8 — Meiers, Michael, University of Waterloo Co-Supervisor Thesis/Project Title: Universal Horizons in Black Holes Present Position: PhD Student, University of California-Davis

2012/8 - 2013/6 — Okoli, Chiamaka, University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Concentration of cosmological haloes Present Position: Deceased

2011/9 - 2013/5 — Yazdi, Yasaman, University of Waterloo Principal Supervisor Thesis/Project Title: A Spacetime Approach to Defining Vacuum States and Entropy Present Position: Avadh Bhatia Postdoctoral Fellow, University of Alberta

2011/8 - 2012/6 — Schuhmann, Robert, University of Waterloo/Perimeter Institute for Principal Supervisor Theoretical Physics Thesis/Project Title: PSI Research Project Present Position: Postdoctoral Research Associate, University of Manchester

2009/8 - 2010/6 — Ponce, Yosdanis, University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Cosmology of a Black Hole Lattice Present Position: Adjunct Faculty in Physics, Miami Dade College, Wolfson Campus

Doctorate

2026/9 - 2030/8 — Alharbi, Ohoud K., University of Waterloo, Principal Supervisor, PhD student

2026/1 — Oberfrank, Robin, Perimeter Institute/University of Waterloo, Co-supervised PhD student

2023/9 - present — Kolonia, Eleanna (Former Research Assistant), PhD student, Swansea University

2023/10 - 2025/8 — Nasiri, Arad, Imperial College London Co-Supervisor Student Degree Expected Date: 2025/8 Thesis/Project Title: Project Supervision: "Stochastic Dark Matter: Covariant Brownian Motion from Planckian Discreteness (arXiv:2409.02188)" Present Position: Postdoctoral Fellow, University of New Brunswick

2023/9 - 2024/12 — Patterson, Everett Avison, University of Waterloo Co-Supervisor Student Degree Expected Date: 2025/12 Thesis/Project Title: Project Supervision: "Accelerated detector in a superposed spacetime (arXiv:2409.12441)" Present Position: PhD Student, University of Waterloo

2023/1 - 2026/12 — Chen, Alice, University of Waterloo and Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Machine Learning and Cosmic Structure Formation Present Position: PhD Student

2022/9 - 2025/8 — Borissova, Johanna, Perimeter Institute for Theoretical Physics/ University Co-Supervisor of Waterloo Thesis/Project Title: Scale-invariance at the core of quantum black holes Present Position: Postdoctoral Fellow, Imperial College London

2022/7 - 2024/12 — Kim, Jungjoon Leo, Queen's University Co-Supervisor Thesis/Project Title: Project Supervision: "A Poisson Log-Normal Framework for Dark Sirens" Present Position: Postdoctoral Fellow, York University

2022/7 - 2022/9	Boyanov Savov, Valentin, Universidad Complutense de Madrid Principal Supervisor Thesis/Project Title: Visiting PhD student: "Aspects of Quantum Black Holes in semiclassical gravity" Present Position: Postdoctoral Researcher, CENTRA and Instituto Superior Tecnico, Lisbon (through April 2026)
2020/9 - 2024/12	Dailey, Conner, University of Waterloo and Perimeter Institute Principal Supervisor Student Degree Expected Date: 2024/12 Thesis/Project Title: On the Initial Boundary Value Problem in Numerical Relativity Present Position: Postdoctoral Researcher, Friedrich Schiller University Jena
2019/8 - 2020/4	Longo, Luis Felipe, University of Sao Paolo, Brazil and Perimeter Institute for Principal Supervisor Theoretical Physics Thesis/Project Title: visiting PhD student: "Echo-Diversity in Binary Black Hole Mergers" Present Position: European Research Council (ERC) Postdoctoral Fellow, Friedrich- Schiller Universitaet, Jena, Germany
2019/2 - 2020/7	Tsuna, Daichi, University of Tokyo Co-Supervisor Thesis/Project Title: visiting PhD student: "Quantum Black Hole Seismology" Present Position: ITC Fellow, Center for Astrophysics, Harvard & Smithsonian
2018/9 - 2019/8	Held, Aaron, University of Heidelberg Principal Supervisor Thesis/Project Title: Visiting PhD student: "Quantum Black Holes in Asymptotic Safety" Present Position: Philippe Meyer Junior Research Chair, Ecole Normale Superieure, Paris
2018/9 - 2019/8	Giri, Utkarsh, University of Waterloo/Perimeter Institute for Theoretical Co-Supervisor Physics Thesis/Project Title: Pressure of haloes in cosmological simulations Present Position: Postdoctoral Scholar Research Associate, California Institute of Technology
2018/9 - 2023/8	Saraswat, Krishan, University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Quantum Aspects of Black Holes: From Microstates to Echoes and Somewhere In-Between Present Position: Postdoctoral Fellow, University of Southern California
2018/1 - 2018/9	Oshita, Naritaka, University of Tokyo Principal Supervisor Thesis/Project Title: visiting PhD student: "Probing microstructure of black hole spacetimes with gravitational wave echoes" Present Position: Assistant Professor, Kyoto University, Yukawa Institute, Kyoto, Japan
2017/9 - 2018/4	Dhumuntarao, Aditya, University of Minnesota Principal Supervisor Thesis/Project Title: visiting NSF PhD student: "Temperature for local observers in Quantum Field Theory in curved spacetime" Present Position: Postdoctoral Researcher, Sandia National Laboratories
2017/9 - 2021/6	Robbins, Matthew, University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Gravitational Waves, Bose-Einstein Condensates, and Modified Gravity Present Position: Teaching Faculty (tenure-track), University of Toronto
2017/9 - 2018/4	Ghersi, Jose Tomas Galvez, Simon Fraser University Principal Supervisor Thesis/Project Title: visiting PhD student: "Temperature for local observers in Quantum Field Theory in curved spacetime" Present Position: Assistant Professor, Universidad de Ingenieria y Tecnologia, Peru
2016/9 - 2020/6	Wang, Qingwen, University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Black Hole Echology Present Position: XVA Quant, Scotiabank
2016/9 - 2018/8	Abedi, Jahed, Sharif University of Technology Co-Supervisor Thesis/Project Title: Black Hole Echoes Present Position: Postdoctoral Fellow, University of Stavanger, Norway
2016/9 - 2017/8	Haghighi, Mohammad Zhoolideh, Sharif University Principal Supervisor Thesis/Project Title: visiting PhD student: "X-ray Cooling Flows and Modified Gravity" Present Position: Assistant Professor, K. N. Toosi University of Technology
2015/9 - 2016/8	Lynch, Morgan, University of Wisconsin-Milwaukee Principal Supervisor Thesis/Project Title: visiting PhD student: "Temperatures of renormalizable quantum field theories in curved spacetime" Present Position: Postdoctoral Researcher, Max Planck Institute for Nuclear Physics
2014/9 - 2018/6	Karami, Mansour, University of Waterloo/Perimeter Institute for Theoretical Co-Supervisor Physics Thesis/Project Title: Dark Matter on Small Scales Present Position: Principal AI Engineer and Quantitative Developer, HOOPP

2014/9 - 2018/6	Altamirano, Natacha, University of Waterloo/Perimeter Institute for Co-Supervisor Theoretical Physics Thesis/Project Title: The quantum and the Gravity: Newtonian and Cosmological Applications Present Position: Director, Data Science, Birch Hill Equity Partners, Toronto, Canada
2014/9 - 2018/6	Okoli, Chiamaka, University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Dark Matter and Neutrinos in the Foggy Universe Present Position: Deceased
2014/4 - 2014/11	Nelson, Elliot, Penn State University Principal Supervisor Thesis/Project Title: Cosmological Non-Constant Problem Present Position: Future Impact Group
2013/9 - 2017/6	Yazdi, Yasaman, University of Waterloo Co-Supervisor Thesis/Project Title: Physics and Astrophysics of Black Holes Present Position: Avadh Bhatia Postdoctoral Fellow, University of Alberta
2013/9 - 2014/8	Narimani, Ali, University of British Columbia Principal Supervisor Thesis/Project Title: visiting PhD student: "How does pressure gravitate?" Present Position: Google
2013/9 - 2014/6	Karami, Mansour, Sharif University of Technology Co-Supervisor Thesis/Project Title: Cosmic Structure Formation, Modified Gravity and Dark Energy, Gravitational Lensing Present Position: Principal AI Engineer and Quantitative Developer, HOOPP
2013/9 - 2017/6	Zwane, Nosiphiwo, University of Waterloo/Perimeter Institute for Theoretical Co-Supervisor Physics Thesis/Project Title: Cosmological Tests of Causal Set Phenomenology Present Position: Lecturer, University of Eswatini
2013/9 - 2014/8	Jeremy Sakstein, University of Cambridge Co-Supervisor Student Degree Start Date: 2010/9 Student Degree Received Date: 2014/7 Thesis/Project Title: visiting PhD student: "Novel Approaches to Modified Gravity" Present Position: Assistant Professor, University of Hawaii
2011/9 - 2017/6	Gould, Elizabeth, University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: A Non-Local Reality: Is there a Phase Uncertainty in Quantum Mechanics? Present Position: Postdoctoral Fellow, Arthur B. McDonald Canadian Astroparticle Physics Research Institute
2011/9 - 2016/6	Saravani, Mehdi, University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Aspects of Non-locality Present Position: Executive Director, Goldman Sachs, United Kingdom
2011/9 - 2014/7	Pourhasan, Razieh, University of Waterloo/Perimeter Institute for Theoretical Co-Supervisor Physics Thesis/Project Title: Adventures in Holography Present Position: Research Assistant, University of Ottawa, Canada
2011/9 - 2015/5	Kamiab, Farbod, University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Neutron Stars, the Exotica - From Modifying General Relativity to Strong Magnetic Fields Present Position: Analytics, Facebook, Dublin, Ireland
2011/1 - 2012/8	Ponce, Yosdanis (Withdrawn), University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Student Degree Start Date: 2011/1 Thesis/Project Title: Black Hole Halo Model/ Holographic Cosmology Present Position: Adjunct Faculty in Physics, Miami Dade College, Wolfson Campus
2010/9 - 2011/9	Baghram, Shant, Sharif University of Technology Co-Supervisor Thesis/Project Title: The Observational Tests of Dark Energy and Modified Gravity Theories Present Position: Assistant Professor, Sharif University, Tehran, Iran
2010/9 - 2011/6	Gryb, Sean, University of Waterloo/Perimeter Institute for Theoretical Co-Supervisor Physics Thesis/Project Title: Shape Dynamics and Mach's Principles: Gravity from Conformal Geometrodynamics Present Position: Research Associate, Philosophy Department, University of Bristol, England
2010/9 - 2014/6	Aslanbeigi, Siavash, University of Waterloo/Perimeter Institute for Principal Supervisor Theoretical Physics Thesis/Project Title: Cosmic Atoms: from Causal Sets to Clusters Present Position: Vice President, Trading Products, BMO Capital Markets, Canada

2008/9 - 2010/6 Prescod-Weinstein, Chanda, University of Waterloo Co-Supervisor Thesis/Project Title: Cosmic Acceleration As Quantum Gravity Phenomenology Present Position: Assistant Professor, University of New Hampshire

2007/9 - 2008/8 Loverde, Marilena, Columbia University Co-Supervisor Thesis/Project Title: Extended Limber Approximation Present Position: Professor and Dr. Ann Nelson Endowed Professor, University of Washington

Post-doctorate

2024/9 - 2026/8 Manita, Yusuke, Perimeter Institute. Former postdoctoral researcher; group appointment completed.

2024/3 - 2024/12 Dunsky, David, New York Univesrity Co-Supervisor Thesis/Project Title: Project Supervision: "Dark Matter Disk Accretion into Black Holes" Present Position: Postdoctoral Fellow, New York Univesrity

2024/3 - 2024/12 Ghalsasi, Akshay, Harvard University Co-Supervisor Thesis/Project Title: Project Supervision: "Dark Matter Disk Accretion onto Black Holes" Present Position: Postdoctoral Fellow, Department of Physics, Harvard University

2024/3 - 2027/3 Faraji, Shokoufe, University of Waterloo Principal Supervisor Thesis/Project Title: Quantum Gravity and Accretion Physics around Black Holes Present Position: Postdoctoral Fellow, Department of Physics and Astronomy, University of Waterloo

2023/3 - 2024/12 Krolewski, Alex, University of Waterloo/ Perimeter Institute for Theoretical Co-Supervisor Physics Thesis/Project Title: Impact of Peculiar Velocities on Color Selection and the Large Scale Structure of Galaxy Surveys Present Position: Research Scientist, California Institute of Technology

2023/1 - 2025/9 Quintin, Jerome, University of Waterloo Co-Supervisor Thesis/Project Title: Quadratic Gravity in Cosmology and Black Holes Present Position: Assistant Professor, ETS, and Visiting Fellow, McGill University

2022/8 - 2024/12 Hassani, Farbod, University of Oslo Principal Supervisor Thesis/Project Title: Project Supervision: "Simulating dynamical friction with cosmic neutrinos" Present Position: Researcher, Institute of Theoretical Astrophysics, University of Oslo

2022/1 - 2024/12 Longo, Luis Felipe, Friedrich-Schiller Universitaet, Jena, Germany Principal Supervisor Thesis/Project Title: Spectrogram Correlated Stacking Present Position: European Research Council (ERC) Postdoctoral Fellow, Friedrich- Schiller Universitaet, Jena, Germany

2021/9 - 2022/8 Muir, Jessica, Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Project supervision: "Search for flattened non-gaussianity" Present Position: Postdoctoral Fellow, Perimeter Institute for Theoretical Physics

2021/3 - 2025/4 Mylova, Maria, Perimeter Institute for Theoretical Physics and IPMU, Japan Principal Supervisor Thesis/Project Title: 1- Non-Gaussian Signatures of a Thermal Big Bang 2- Effective Cuscuton Theory Present Position: Institute for Basic Science (IBS), South Korea

2020/10 - 2024/12 Nikakhtar, Farnik, Yale University Co-Supervisor Thesis/Project Title: Project Supervision: "Action-Angle Space Clustering" Present Position: YCAA Postdoctoral Fellow, Yale University

2020/3 - 2024/3 Dey, Ramit, Western University, Ontario, Canada Co-Supervisor Thesis/Project Title: 1- Spectrogram correlated stacking: A novel time-frequency domain analysis of the Stochastic Gravitational Wave Background 2- Black Hole Echoes Present Position: Postdoctoral Fellow, Western University, and Associate Postdoctoral Researcher, Perimeter Institute

2020/2 - 2020/6 Shoshany, Barak, Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Bayesian Search for Black Hole Echoes Present Position: Associate Professor, Brock University

2019/4 - 2021/3 Oshita, Naritaka, Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Echoes from Quantum Black Holes Present Position: Assistant Professor, Kyoto University, Yukawa Institute, Kyoto

2019/1 - 2023/12 Abedi, Jahed, University of Stavanger, Norway Principal Supervisor Thesis/Project Title: Search for Black Hole Echoes in Gravitational Wave Observations Present Position: Assistant Professor, Beijing Institute for Mathematical Sciences and Applications (BIMSA)

2015/9 - 2018/6 Nelson, Elliot, Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Quantum Decoherence during Inflation from Gravitational Nonlinearities Present Position: Future Impact Group

2013/9 - 2017/8 Guariento, Daniel, Universidade de Sao Paulo/Perimeter Institute for Principal Supervisor Theoretical Physics Thesis/Project Title: Cosmological Black Holes from Self-Gravitating Fields Present Position: Professor, Conestoga College, Canada

2013/9 - 2014/8 Scrimgeour, Morag, University of Waterloo Principal Supervisor Thesis/Project Title: Mass of Neutrinos from Neutrino Wakes Present Position: Data Scientist, Square, San Francisco CA

2012/9 - 2015/8 Marsh, David, Perimeter Institute for Theoretical Physics Thesis/Project Title: Structure of Dark Matter Haloes Present Position: Sofja Kovaleskaja Fellow, University of Goettingen

2012/9 - 2016/8 Kubiznak, David, Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Black Hole Chemistry Present Position: Senior Fellow, Perimeter Scholar International program

2012/9 - 2016/8 Hinterbichler, Kurt, Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Consistent modifications of Einstein gravity Present Position: Assistant Professor, Case-Western Reserve University

2012/9 - 2015/8 Kobayashi, Takeshi, Perimeter Institute for Theoretical Physics/Canadian Principal Supervisor Institute for Theoretical Astrophysics Thesis/Project Title: Primordial Magnetic Fields from the Post-Inflationary Universe Present Position: Associate Professor, SISSA (Scuola Internazionale Superiore di Studi Avanzati)

2011/9 - 2013/8 Lavaux, Guilhem, University of Waterloo/Perimeter Institute-CITA National Co-Supervisor Fellow Thesis/Project Title: A New Light on Missing Baryons Present Position: CNRS Permanent Position, L'Institut d'Astrophysique de Paris, France

2011/9 - 2012/8 Baghram, Shant, University of Waterloo/Perimeter Institute for Theoretical Principal Supervisor Physics Thesis/Project Title: Prospects for Detecting Dark Matter Halo Substructure with Pulsar Timing Present Position: Professor st Sharif University, Sharif University, Tehran, Iran

2010/9 - 2013/8 Zavala, Jesus, University of Waterloo - CITA National Fellow Principal Supervisor Thesis/Project Title: Clustering in the Phase Space of Dark Matter Haloes Present Position: Professor at U-Iceland, University of Iceland

2010/9 - 2012/8 Johnson, Matthew, Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Determining the Outcome of Cosmic Bubble Collisions in Full General Relativity Present Position: Associate Professor at York University, Associate Faculty at Perimeter Institute

2009/9 - 2011/8 Shandera, Sarah, Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Breakdown of Semiclassical Methods in de Sitter Space Present Position: Associate Professor, Penn-State University

2009/9 - 2013/8 Erickcek, Adrienne, Perimeter Institute for Theoretical Physics/Canadian Principal Supervisor Institute for Theoretical Astrophysics Thesis/Project Title: Catastrophic Consequences of Kicking the Chameleon Present Position: Assistant Professor, University of North Carolina-Chapel Hill

Research Associate

2021/9 - 2022/5 Mukherjee, Suvodip, Perimeter Institute for Theoretical Physics Principal Supervisor Thesis/Project Title: Spectrogram correlated stacking: A novel time-frequency domain analysis of the Stochastic Gravitational Wave Background Present Position: Reader, Tata Institute of Fundamental Research, India

Staff Supervision

Number of Visiting Researchers: 1

Event Administration

2020/7 - 2024/12	Member of the Organizing Committee, "Copernicus Weekly Webinar Series" focusing on cosmology and high energy physics https://indico.cern.ch/event/936284/overview, Seminar, 2020/7 - 2024/12
2023/3 - 2024/7	Local Organizing Committee (Chair), "50 Years of Horndeski Gravity: Exploring Modified Gravity" University of Waterloo and Perimeter Institute for Theoretical Physics, Conference, 2024/7 - 2024/7
2019/10 - 2024/6	Scientific Advisory Committee member, "[RN50] The Event Horizon and Beyond - Celebrating 50 Years of Narayan" Boston Museum of Science, Boston, MA https://www.cfa.harvard.edu/events/rn50-celebrating-50-years-narayan, Conference, 2024/6 - 2024/6
2023/11 - 2024/5	Local Organizing Committee (chair), "McNamara@65: Understanding Feedback in Galaxies and Clusters" University of Waterloo, Conference, 2024/5 - 2024/5
2023/11 - 2024/4	Scientific Organzing Committee Member, "Cosmological Frontiers in Fundamental Physics" (University of Edinburgh), Conference, 2024/4 - 2024/4
2022/10 - 2023/5	Organizing Committee (Chair), "Quantum Spacetime in the Cosmos: From Conception to Reality" Perimeter Institute, Conference, 2023/5 - 2023/5
2020/8 - 2021/5	Scientific Organzing Committee Member, "Cosmological Frontiers in Fundamental Physics" (Virtual at APC, Paris), Conference, 2021/5 - 2021/5
2020/2 - 2020/2	Organizer, "Echoes in Southern Ontario" Perimeter Institute, Workshop, 2020/2 - 2020/2
2019/5 - 2019/11	Co-organizer, Everpresent Lambda: Theory meets Observations, Conference, 2019/11 - 2019/11
2019/3 - 2019/9	Lead Organizer, "Cosmological Frontiers in Fundamental Physics 2019" Perimeter Institute, Conference, 2019/9 - 2019/9
2019/2 - 2019/7	Chairperson of Parallel Session, 22nd International Conference on General Relativity and Gravitation, Conference, 2019/7 - 2019/7
2017/9 - 2019/4	Chair, Astrophysics Seminar Chair, University of Waterloo, Seminar, 2017/9 - 2019/4
2019/1 - 2019/2	Scientific Organizing Committee, Tehran Meeting on Cosmology at the Crossroads (Virtual), Conference, 2019/2 - 2019/2
2017/9 - 2018/12	Chair, Physics 10 Undergraduate Lectures, University of Waterloo, Course, 2017/9 - 2018/12
2018/1 - 2018/6	Co-organizer, Asymptotic Safety in a Dark Universe, Conference, 2018/6 - 2018/6
2017/5 - 2017/11	Co-organizer, Quantum Black Holes in the Sky?, Conference, 2017/11 - 2017/11
2016/1 - 2016/7	Co-organizer, Philosophy of Cosmology, Conference, 2016/7 - 2016/7
2015/6 - 2015/12	Co-organizer, Mann Fest, Conference, 2015/12 - 2015/12
2015/2 - 2015/8	Co-organizer, Cosmic Flows (and Other Novelties on Large Scales), Conference, 2015/8 - 2015/8
2014/3 - 2014/4	Co-organizer, Implications of BICEP2, Workshop, 2014/4 - 2014/4

2011/5 - 2012/5	Co-organizer, Conformal Nature of the Universe, Conference, 2012/5 - 2012/5
2011/10 - 2012/4	Co-organizer, Theoretical Physics Workshop, Workshop, 2012/4 - 2012/4
2010/9 - 2011/9	Co-organizer, Unravelling Dark Matter 2011, Conference, 2011/9 - 2011/9
2009/12 - 2010/6	Co-organizer, Cosmological Frontiers of Fundamental Physics, Workshop, 2010/6 - 2010/6
2008/9 - 2009/8	Chair, Cosmology Seminar Series, Seminar, 2008/9 - 2009/8
2008/11 - 2009/5	Co-organizer, New Prospects for Solving the Cosmological Constant Problem, Workshop,
2009/5	- 2009/5
2008/5 - 2009/5	Co-organizer, Sunyaev-Zel'dovich Universe and the Future of Cluster Cosmology, Conference, 2009/4 - 2009/5
2008/10 - 2008/12	Co-organizer, Young Researchers Conference, Conference, 2008/12 - 2008/12
2008/4 - 2008/10	Co-organizer, PI/CITA Cosmology Mini-Workshop, Workshop, 2008/10 - 2008/10
2007/12 - 2008/6	Co-organizer, Small Scale Structure of Dark Matter, Workshop, 2008/6 - 2008/6
2008/1 - 2008/3	Co-organizer, Origins and Observations of Primordial Non-Gaussianity, Workshop, 2008/3 - 2008/3

Editorial Activities

2014/1 - 2020/8	Scientific Editor, Research in Astronomy and Astrophysics, Journal
2012/9 - 2013/8	Editorial Board Member, Phys13 News, Newsletter
2010/9 - 2011/8	Editorial Board Member, Phys13 News, Newsletter

Graduate Examination Activities

2024/9 - 2024/9	PhD External Examiner, Danieli, Valentina, Astroparticle Physics, Scuola Inter. Superiore di Studi Avanzati
2024/9 - 2024/9	PhD External Examiner, Velluci, Vania, Astroparticle Physics, Scuola Inter. Superiore di Studi Avanzati

Community and Volunteer Activities

2009/9	Speaker/Educator, Perimeter Institute for Theoretical Physics I am a regular contributor to outreach events at Perimeter Institute. My responsibilities include giving lectures, participating in Q&A's, and discussions with primary and high school students, as well as high school teachers.

Knowledge and Technology Translation

2026/3 - present	Consultant, Aligned. Activity Description: Testing frontier AI models in scientific research.
2020/2 - 2023/5	Creator, Research Uptake Strategies Target Stakeholder: General Public Outcome / Deliverable: Based on our study of North American demographic, mobility, climate, and COVID-19 data, we provided a predictive model for growth and mitigation strategies for each US and Canadian city and county. The models were publicly available in the form of an online portal, during April 2021- May 2023. References / Citations / Web Sites: Canada: http://mylocalcovid.uwaterloo.ca [decommissioned in May 2023] US: https://wolfr.am/COVID19Dash
2020/4 - 2021/10	Scientific Advisor, Consultation Service Group/Organization/Business Serviced: Public Broadcasting Company (PBS) Target Stakeholder: General Public Outcome / Deliverable: NOVA Universe Revealed: Big Bang, Venture back to the birth of space and time as NOVA probes the mystery of the Big Bang. Premiered: 11/24/21 at 9pm EST Runtime: 53:16, PBS documentary Evidence of Uptake/Impact: While the specific viewership for this program is not publicly released, PBS is the premier public national broadcaster in the US. Each month, approximately 100 million people watch their local PBS stations. References / Citations / Web Sites: https://www.pbs.org/wgbh/nova/video/nova-universerevealed-big-bang/ Activity Description: I acted as the scientific advisor for the PBS NOVA documentary: "NOVA Universe Revealed: Big Bang", and helped with creating a scientifically accurate portrait of the Big Bang

International Collaboration Activities

2019/8 - 2029/8	Member of Laser Interferometer Space Antenna (LISA) consortium, Canada The Laser Interferometer Space Antenna (LISA) is a planned space-based observatory designed to detect and study gravitational waves. As a member of the LISA consortium, I am contributing to the agenda and ideas for the "Fundamental Physics" and "Waveform Modelling" working groups.

Committee Memberships

2024/7 - 2025/7	Committee Member, Seminars and Colloquia Committee, Perimeter Institute for Theoretical Physics
2020/9 - 2024/8	Committee Member, Undergraduate Experience Committee, University of Waterloo Organize and coordinate events to improve student experience
2023/9 - 2024/7	Committee Member, Postdoctoral Search Committee, Perimeter Institute for Theoretical Physics
2021/9 - 2023/8	Chair, PhD Student Student Affairs, Perimeter Institute for Theoretical Physics
2013/1 - 2022/7	Committee Member, Early Researcher Award Grant Review Panel, Ministry of Research and Innovation, Government of Ontario I served on the review panel for most years, during this period.
2019/9 - 2020/8	Committee Member, Emmy Noether Fellowship Search Committee, Perimeter Institute for Theoretical Physics
2019/7 - 2019/10	Chair, "Astrophysics Theory Panel" grant review committee, National Aeronautics and Space Administration (NASA) Chair of a review panel for Theoretical Astrophysics grants
2014/8 - 2017/8	Chair, Colloquium Committee, Department of Physics and Astronomy, University of Waterloo
2013/9 - 2014/8	Committee Member, Co-op Report Marking Committee, University of Waterloo Ongoing Membership
2010/9 - 2012/8	Committee Member, Seminar/Colloquium Committee, University of Waterloo
2010/9 - 2011/8	Committee Member, SIN Exam Committee, University of Waterloo

2010/9 - 2011/8 Committee Member, PI/UW Astronomy Faculty Search Committee, University of Waterloo

2008/9 - 2010/8 Committee Member, Postdoctoral Recruitment Committee, Perimeter Institute for Theoretical Physics

Other Memberships

2021/1 Member, Canadian Institute for Theoretical Astrophysics, Incorporated (CITA, Inc) The Canadian Institute for Theoretical Astrophysics (CITA), located at the University of Toronto, is a leading research centre dedicated to exploring fundamental questions in theoretical astrophysics. CITA's research spans areas such as cosmology, galaxy formation, black holes, high-energy astrophysics, and planetary science. As a member of CITA Inc., I collaborate with other members across the country to advance theoretical astrophysics. As part of this organization, I contribute to recruiting top talent to Canada and mentoring highly qualified personnel, fostering the next generation of astrophysicists in this dynamic field.

2015/1 Ordinary Member, Canadian Astronomical Society (CASCA) CASCA is a professional society for Canadian astronomers and researchers involved in the fields of astronomy, astrophysics, and related disciplines. The primary goals of CASCA are to promote the advancement of astronomical research, education, and outreach in Canada.

2009/9 - 2014/8 Faculty of Science representative at the Faculty of Math Council, University of Waterloo Ongoing appointment

Presentations

1 (2026). *Scrambled Spacetime: Qubits, black holes, and the speed of causality.* Qiskit Fall Fest 2026, University of Waterloo, B1 271, 5 November 2026. Slides.

2 (2026). *Quantum Gravity in the Wild: The Big Bang, Black Holes, and a Cosmic Glitch.* Physics and Astronomy Colloquium, University of New Mexico, Albuquerque, United States of America, 23 October 2026. Event details and recording link.

3 (2026). *Making Quantum Gravity Observable: From the Big Bang to Black Holes and Cosmic Horizons.* Invited talk, From Quantum Gravity to Cosmology, Kavli IPMU, The University of Tokyo, Kashiwa, Japan, 1 October 2026. Event details.

4 (2026). *Quantum Gravity in the Wild: Black Hole Echoes, the Big Bang, and a Cosmic Glitch.* Joint Colloquium, ITP/CEICO, Prague, 21 August 2026.

5 (2026). *The Natural Log Project: A LISA benchmark for entropy-scale quantum black-hole signatures.* LISA Consortium status update (virtual), 27 July 2026.

6 (2026). Quantum Gravity in the Wild: Testing Horizon-Scale Physics with Gravitational Waves and Cosmology. Recent Development of Quantum Gravity and Applications to Cosmology and Black Hole Physics, Hangzhou, China

7 (2026). Quadratic Gravity panel contribution. Recent Development of Quantum Gravity and Applications to Cosmology and Black Hole Physics, Hangzhou, China

8 (2026). Massive Neutrinos. Perimeter Institute for Theoretical Physics, Waterloo, ON, Canada

9 (2026). *Battle of the Big Bang.* Howard University, Washington, DC, United States of America

10 (2026). *Battle of the Big Bang.* Kitchener Public Library, Kitchener, ON, Canada

11 (2026). *Opening Remarks.* Cosmological Frontiers in Fundamental Physics 2026, Perimeter Institute, Waterloo, ON, Canada, 27 April 2026. PIRSA:26040060.

12 (2025). *Quantum Gravity in the Wild.* Gravity, Particles, and Information Conference (MannFest at 70), University of Waterloo, Waterloo, ON, Canada, 17 December 2025. Conference schedule.

13 (2025). *Quantum Gravity in the Wild*. Rethinking Cosmology, Raman Research Institute, Bengaluru, India, 11 December 2025. Speaker record.

14 (2025). *Quantum Black Holes in the Sky*. 50 Years of the Black Hole Information Paradox, Simons Center for Geometry and Physics, Stony Brook, NY, United States of America, 3 November 2025. Speaker record.

15 (2025). *Battle of the Big Bang: New Tales of Our Cosmic Origins*. Physics Colloquium, University of Edinburgh, Edinburgh, United Kingdom, 24 October 2025. Event details.

16 (2025). *UV-Completing the Universe in Quadratic Gravity*. Dublin Institute for Advanced Studies, Dublin, Ireland, 23 October 2025. Speaker record.

17 (2025). *Battle of the Big Bang*. Secular AZ virtual seminar series, Arizona, United States of America, 8 October 2025. Speaker record.

18 (2025). *Battle of the Big Bang*. Public Lecture, Royal Institution of Great Britain, London, United Kingdom, 28 June 2025. Recording.

19 (2025). *UV-Completing the Universe in Quadratic Gravity*. King's College London, London, United Kingdom, 27 June 2025. Speaker record.

20 (2025). *UV-Completing the Universe in Quadratic Gravity*. Imperial College London, London, United Kingdom, 26 June 2025. Speaker record.

21 (2025). *Tempus Discretum: On the Nature of Cosmic Time*. Lee's Fest: Quantum Gravity and the Nature of Time, Perimeter Institute, Waterloo, ON, Canada, 4 June 2025. PIRSA:25060046.

22 (2025). *Battle of the Big Bang: The New Tales of Our Cosmic Origins*. Public Lecture, Perimeter Institute, Waterloo, ON, Canada, 4 June 2025. PIRSA:25060061.

23 (2025). *Battle of the Big Bang*. Ottawa Intellectual Dialogue, 15 May 2025. Speaker record.

24 (2025). Quantum Black Holes & the Conundrum of Cosmic Causality. Physics Colloquium, Arizona State University, Phoenix, AZ, United States of America

25 (2025). Battle of the Big Bang: How the Universe Began (and Why It's a Big Mystery!). Physics Literacy for Kids (Zoom talk),

26 (2025). Copenhagen Survey on Black Holes and Fundamental Physics. Physics Colloquium (virtual), Tehran, Iran

27 (2025). Quantum Black Holes & the Conundrum of Cosmic Causality. Physics & Astronomy Colloquium, McMaster University, Hamilton, ON, Canada

28 (2025). Quantum Causality & the Crisis in Physics: Black Holes, the Big Bang, and Beyond. Sydney CPPC virtual seminar, Sydney, Australia

29 (2024). *Quantum Black Holes & the Conundrum of Cosmic Causality*. University of Chinese Academy of Sciences, Beijing, China, 5 November 2024. Speaker record.

30 (2024). *The Big Picture*. University College London, London, United Kingdom, 15 April 2024. Speaker record.

31 w/ Stefano Liberati (SISSA). (2024). Quantum Black Holes (concluding panel discussion, and wrap-up). Black Holes inside and out, Niels Bohr Institute, Copenhagen, Denmark

32 (2024). Precision Quantum Cosmology: Past, Present, and Future. Gravity and Cosmology 2024, Yukawa Institute for Theoretical Physics, Kyoto University, Kyoto, Japan

33 (2024). Quantum Spacetime in the Sky: from the horizon to horizon. Heising-Simons Foundation (HSF) Collaboration on Quantum Gravity and Its Observational Signatures (QuRIOS) Journal Club (virtual), Waterloo, ON, Canada

34 (2024). Quantum Black Holes in the Sky. The Event Horizon and Beyond, Celebrating 50 years of Narayan, Boston, MA, United States of America

35 (2024). Quantum Black Holes & the Conundrum of Cosmic Causality. Physics Colloquium, International Center for Theoretical Physics Asia-Pacific (ICTP-AP), Beijing, China

36 (2024). Quantum Spacetime in the Sky: from the horizon to the vacuum. Bottom-Up Cross-Cutting Workshop "JENAS Initiative: Gravitational Wave Probes of Fundamental Physics" 12–16 Feb 2024 Sapienza University of Rome, Rome, Italy

37 (2024). Quantum Black Holes & the Conundrum of Cosmic Causality. Physics Colloquium, Institute of High Energy Physics, Beijing, China

38 (2024). Quantum Spacetime in the Sky: from the horizon to horizon. International Congress of Basic Science 2, Beijing Institute for Mathematical Science and Applications (BIMSA), Beijing, China

39 (2024). Imaging Quantum Black Holes & (Towards) Optimal Estimators of Stochastic Gravitational Wave Background. The 1st Tsinghua Relativity Workshop, Beijing, China

40 (2023). *Welcome and Opening Remarks*. Quantum Spacetime in the Cosmos: From Conception to Reality, Perimeter Institute, Waterloo, ON, Canada, 8 May 2023. PIRSA:23050146.

41 (2023). The Big Picture: New Vistas in Gravitational Wave Astronomy. Canadian Astronomical Society (CASCA) Annual General Meeting, Penticton, BC, Canada

42 (2023). Quantum Spacetime and the Cosmic Endgame. Physics Colloquium, Simon Fraser University, Burnaby, Canada

43 (2023). A Tale of Stars and Sirens. The Past, Present and Future of Astronomical Surveys - A Celebration of the Scientific Careers of Michael Strauss and Robert Lupton, Princeton University, Princeton, NJ, United States of America

44 (2023). Quantum Gravity and the Cosmos. Testing Aspects of General Relativity-II (virtual), Waterloo, ON, Canada

45 (2023). The paradox of human and the cosmos. Ottawa Intellectual Discussions (Virtual), Waterloo, ON, Canada

46 (2023). The Endgame: Quantum Spacetime and the Cosmos. Physics Colloquium, University of British Columbia, Vancouver, BC, Canada

47 (2023). The Big Picture. DE Shaw Company (virtual seminar), Waterloo, ON, Canada

48 (2023). Quantum Spacetime in the Sky: from Conception to Reality. ISRA 2023: The International Symposium on Recent Developments in Relativistic Astrophysics, Gangtok, India

49 (2023). Into the Unknown: A Journey on the Dark Side. Virtual Public Lecture for Lord Byng high school Astronomy Club (Vancouver, BC), Waterloo, ON, Canada

50 (2023). From Black Holes to the Big Bang: Humanity's Cosmic Paradox. Public Lecture at Kitchener Public Library, Kitchener, ON, Canada

51 (2023). Spectrogram correlated stacking: A novel time-frequency domain analysis of the Stochastic Gravitational Wave Background. Cosmology and Gravitation Seminar, Perimeter Institute, Waterloo, ON, Canada

52 (2023). Quantum Precision Cosmology. Testing Gravity 2023, Vancouver, BC, Canada

53 (2023). The War Within: Humanity's Cosmic Paradox. Physics 10 seminar series, University of Waterloo, Waterloo, ON, Canada

54 (2022). Einstein vs Einstein. Testing Aspects of General Relativity (Joint Canadian/Indian virtual conference), Waterloo, Canada

55 (2022). Einstein vs Einstein: The Great War. Physics Colloquium Institute for Research in Fundamental Sciences (IPM), (remote presentation), Tehran, Iran

56 (2022). Mapping the Big Bang. Remote Public Talk at Farzanegan High School, Tehran (virtual talk in Persian), Tehran, Iran

57 (2022). Stimulating Hawking Radiation of Gravitational waves: from theory to observation. New horizons for (no-)horizon physics: from gauge to gravity and back, Galileo Galilei Institute, Florence, Italy

58 (2022). Einstein vs Einstein: The Great War. Cosmo Coffee Seminar Series, Theory Group, the European Organization for Nuclear Research (CERN) (remote presentation), Geneva, Switzerland

59 (2022). Stimulating Hawking Radiation of Gravitational Waves. Quantum Simulators of Fundamental Physics (virtual seminar series), Waterloo, Canada

60 (2022). Quantum Black Holes: from holography to scrambling and gravitational wave astronomy. The 1st International Conference of Holography and its Applications, Damghan University (remote presentation), Damghan, Iran

61 (2022). Fundamental Physics vs. Fundamentalism. History, Philosophy, and Sociology of Cosmology and Astroparticle Physics Conference, Bonn, Germany

62 (2021). Cosmology of the Pandemic. Journal-club Galaxies, Institut d'Astrophyisque de Paris (remote talk), Paris, France

63 (2021). Einstein vs Einstein: The Great War. Gravity Seminar Series, University of Southampton (remote presentation), Southampton, United Kingdom

64 (2021). Black Holes, Unscripted. Copernicus Webinar Series (virtual presentation), Waterloo, Canada

65 (2021). On the origin of the LIGO "mystery" noise and the high energy particle physics desert. Sixteenth Marcel Grossmann Meeting (virtual), Waterloo, ON, Canada

66 (2021). Physics and Astrophysics with Gravitational Waves. 14th Iranian national conference on Astronomy and Astrophysics, Semnan University (Virtual public talk in Persian), Semnan, Iran

67 (2021). Big Bang: Fact vs Fiction. The 28th IPM Physics Spring Conference, Tehran, Iran (remote presentation), Tehran, Iran

68 (2021). Cosmology and the Pandemic. Virtual conference on Gravity and Cosmology, Shahid Beheshti University (Remote keynote presentation in Persian), Tehran, Iran

69 (2021). A New Way to Measure Mass. Streams 21 (virtual conference), Waterloo, Canada

70 (2021). Cosmology: Triumphs, Trials, and Tribulations. Physics Colloquium, Tarbiat Modares University (remote presentation), Tehran, Iran

71 (2021). From Cosmos to the Pandemic. 37th Science and Technology Conference, Allameh Helli High School (virtual talk in Persian), Tehran, Iran

72 (2021). Into the Unknown. Physics Colloquium, University of Northern Iowa (remote), Cedar Falls, IA, United States of America

73 (2021). Black Holes, Unscripted. Black-Hole Microstructure, virtual conference organized by U-Saclay, France (remote presentation), Saclay, France

74 (2021). From Cosmology to the Pandemic. Sharif University of Technology Alumni Associate Webinar Series (virtual talk in Persian), Waterloo, ON, Canada

75 (2021). On Nothing!. Alternative Gravities and Fundamental Cosmology 2021 (virtual conference, organized by University of Szczecin, Poland), Szczecin, Poland

76 (2021). On the origin of the LIGO "mystery" noise and the high energy particle physics desert. The XXVIII International Conference on Supersymmetry and Unification of Fundamental Interactions (SUSY 2021, virtual), Waterloo, ON, Canada

77 (2021). From the Cosmos to the Pandemic: A Dive into the Unknown. 26th Special Physics School, Institute for Advanced Studies in Basic Sciences (remote talk in Persian), Zanjan, Iran

78 (2020). Probing Physics Beyond the Standard Model with the Gravitational Wave "Mystery" Noise. Quantum Spacetime and the Renormalization Group (organized by CP3 Origins, Denmark, virtual conference), Waterloo, ON, Canada

79 (2020). Local drivers of growth and containment for COVID-19 pandemic. Seminar Series on the Mathematical Modelling of COVID-19, Fields Institute for Research in Mathematical Sciences, Toronto, Canada

80 (2020). A New Way to Measure Mass. CANadian Virtual Astronomy Seminar (CANVAS) (virtual presentation), Waterloo, Canada

81 (2020). On the Edge of the Abyss. Virtual Seminar, McGill Space Institute, Montreal, QC, Canada

82 (2020). A New Way to Measure Mass. Galactic Dynamics Journal Club, University of Pennsylvania (remote presentation), Philadelphia, PA, United States of America

83 (2020). Quantum Black Hole Seismology. The 13th International LISA Symposium (virtual conference, prerecorded talk), Waterloo, Canada

84 (2020). Quantum Black Hole Seismology. 23rd meeting in Astronomical Research, Zanjan, Iran (remote presentation), Zanjan, Iran

85 (2020). A novel predictive framework for the novel coronavirus. Centre for Disease Modelling, Canada- China Distinguished Lecture, Canada

86 (2020). Quantum Black Holes in the Sky. Quantum Gravity and the Renormalization Group, Odense, Denmark

87 (2020). Managing the COVID-19 Pandemic across Geography and Demography. seminar for the Google COVID-19 data working group (virtual), Waterloo, ON, Canada

88 (2020). On the secrets of black holes, worm holes, white holes, and the firewalls. Remote public talk in Persian for Astronomy (Nojoom) Magazine, Tehran, Iran

89 (2020). Quantum Black Hole Seismology. Astrophysics Seminar Series, Bar-Ilan University, Israel (remote presentation), Israel

90 (2020). Quantum Black Hole Seismology. High Energy Physics (HEP) seminar series, school of physics, Institute for Research in Fundamental Sciences (IPM), Tehran (remote presentation), Tehran, Iran

91 (2020). Managing the COVID-19 Pandemic across Geography and Demography. Seminar Series, Astronomy School, Institute for Research in Fundamental Sciences (IPM), Tehran (remote, tehran, Iran

92 (2020). A New Way to Measure Mass. Astrophysics Seminar, Sharif University (remote presentation), Tehran, Iran

93 (2020). The invisible 99 percent. Physics Club (remote public talk in Persian), Tehran, Iran

94 (2020). Managing the COVID-19 Pandemic across Geography and Demography. Colloquium, Perimeter Institute, Waterloo, ON, Canada

95 (2020). Hubble Tension and an Everpresent Lambda. 3rd Online Workshop on Quantum Gravity, “Quantum Gravity meets dark energy” (virtual conference), waterloo, Canada

96 *Aslanbeigi, S; *Saravani, M. (2019). Dark Matter from Quantum Gravity. Everpresent Lambda: Theory Meets Observations, Perimeter Institute for Theoretical Physics, Waterloo, ON, Canada

97 (2019). Dark Energy [On Nothing]. CosmoCruise 2019, Venice, Italy

98 (2019). A Tale of Big Bang. Milky Way Persian Science Channel, Iran (virtual), Waterloo, ON, Canada

99 (2019). Quantum Black Holes in the Sky. Testing Gravity 2019, Vancouver, BC, Canada

100 (2019). Cosmological non-Constant (CnC) Problem. 5th Bahamas Advanced Study Institute and Conferences 2019 (BASIC 2019) (remote presentation), Bahamas

101 (2019). Quantum Black Holes in the Sky. Testing Gravity 2019, Vancouver, Canada

102 (2019). On Nothing!. Physics Colloquium at Grand Valley State University, Grand Rapids, MI, United States of America

103 (2018). Quantum Black Holes in the Sky: From Quantum Gravity to Astrophysics and Cosmology. Solvay-Perimeter-APC Workshop on "Cosmological Frontiers in Fundamental Physics" - Solvay Institute, Brussels, Belgium

104 (2018). Quantum Black Holes in the Sky: from Quantum Gravity to Astrophysics and Cosmology. Cosmology and Gravitation Seminar Series, Perimeter Institute, Waterloo, ON, Canada

105 (2018). Seeing through Simulation Blindspots. Rotman Summer Institute on Philosophy of Cosmology, Goderich, Canada

106 (2018). Reviving Aether in Einstein's Universe: Trials, Tribulations, and Triumphs. Lecture for undergraduate Cosmology and Gravitation class at Western University, London, ON, Canada

107 (2018). Seeing Cosmic Nanostructure: From Plasma to Dark Matter. Seminar - Canadian Institute for Theoretical Astrophysics, University of Toronto, Toronto, Canada

108 (2018). Quantum Black Holes in the Sky: from Quantum Gravity to Astrophysics and Cosmology. Physics and Astronomy Colloquium, Western University, London, ON, Canada

109 (2018). Quantum Black Holes in the Sky: from Quantum Gravity to Astrophysics and Cosmology. Theoretical High Energy Physics Seminar Series, McGill University, Montreal, QC, Canada

110 (2018). Adios, Lorentz! Hola, Kuhn!. Rotman Summer Institute on Philosophy of Cosmology, Goderich, Canada

111 (2018). Cosmological non-Constant Problem. Asymptotic Safety in a Dark Universe, Perimeter Institute, Waterloo, ON, Canada

112 (2018). Quantum Black Holes in the Sky: From Quantum Gravity to Astrophysics and Cosmology. HEP Seminar - McGill University, Montreal, Canada

113 (2018). The Other 99 Percent: A Dark Cosmos. Keynote lecture for high school teachers, Perimeter Institute, Waterloo, ON, Canada

114 (2018). Quantum Black Holes in the Sky: from Quantum Gravity to Astrophysics and Cosmology. Cosmology and Gravitation Seminar, Perimeter Institute, Waterloo, ON, Canada

115 (2017). Echoes from the Abyss. Testing Gravity 2017, Vancouver, Canada

116 (2017). Reflections on Spacetime. Cosmology and the Future of Spacetime - Western University, London, Canada

117 (2017). Echoes from the Abyss. Seminar - Department of Applied Mathematics, University of Waterloo, Waterloo, Canada

118 (2017). Quantum Black Holes in the Sky. Colloquium - University of Kansas, Lawrence, Kansas, United States of America

119 (2017). Quantum Black Holes in the Sky. Colloquium - Department of Physics, Ohio State University, Columbus, United States of America

120 (2017). Quantum Black Holes in the Sky - Plenary Lecture. CosPA 2017, Kyoto, Japan

121 (2017). How to turn Quantum Gravity into Science?. Probing the Spacetime Fabric: From Concepts to Phenomenology - SISSA, Trieste, Italy

122 (2017). Echoes from the Abyss. Physics and Astrophysics at the eXtreme (PAX), Amsterdam, Netherlands

123 (2017). Seeing Dark in the Extremes: From Nanostructure, to Infinity and Beyond. CPARC Colloquium - Queen's University, Kingston, Canada

124 (2016). The Many Faces of the Cosmological Constant Problem. Experimental Search for Quantum Gravity - Frankfurt Institute for Advanced Studies, Frankfurt, Germany

125 (2016). Cosmological Non-Constant Problem: Astrophysical Limits of Particle Physics. HET-Discovery Seminar - Niels Bohr Institute, University of Copenhagen, Copenhagen, Denmark

126 (2016). E.F.T. is Evil!. Unifying Tests of General Relativity - Walter Burke Institute for Theoretical Physics, Caltech, Pasadena, United States of America

127 (2016). Accretion in Radiative Equipartition. Seminar - Dark Cosmology Centre, University of Copenhagen, Copenhagen, Denmark

128 (2015). (Real) Cosmology with Quark-Gluon Plasmas. Cosmology/HEP Seminar - Brookhaven National Laboratory, Upton, New York, United States of America

129 (2015). Pulsar Timing Constraints on Physics Beyond Standard Model. Mann Fest - University of Waterloo, Waterloo, Canada

130 (2015). Cosmological (Non-)Constant Problem. Testing Gravity 2015, Vancouver, Canada

131 (2015). First Erlangen Workshop on Quantum Gravity and Cosmology. High Energy Physics Seminar - University of Toronto, Toronto, Canada

132 (2015). Cosmological Non-Constant Problem: The Case for TeV Scale Quantum Gravity. First Erlangen Workshop on Quantum Gravity and Cosmology, Erlangen, Germany

133 (2015). Alice in the (Relativistic) Wonderland. Colloquium - Department of Mathematics, Western University, London, Canada

134 (2014). Probing Dark Matter Nanostructure. Mitchell Workshop on Collider and Dark Matter Physics, College Station, United States of America

135 (2014). From Black Holes to Big Bang, and Back!. Cosmology Seminar - Perimeter Institute for Theoretical Physics, Waterloo, Canada

136 (2014). Cosmological (Non)-Constant Problem. Large Angle Anomalies: from a posteriori to a priori - Case-Western Reserve University, Cleveland, United States of America

137 (2014). An Introduction to the Universe - Plenary Talk. Sharif University Alumni Conference, Tehran, Iran

138 (2014). From Black Holes to Big Bang, and Back!. Colloquium - Department of Physics, Isfahan University of Technology, Isfahan, Iran

139 (2014). Probing Dark Matter Nanostructure. Cosmology Seminar - Department of Physics, Sharif University of Technology, Tehran, Iran

140 (2014). Dark Matter Nanostructure. Astrophysics Seminar - Department of Physics and Astronomy, University of Waterloo, Waterloo, Canada

141 (2014). From Black Holes to Big Bang, and Back!. Invited Colloquium - School of Astronomy, Institute for Research in Fundamental Sciences (IPM), Tehran, Iran

142 (2014). An Introduction to the Universe, Q&A. Zaferanieh Amateur Observatory, Tehran, Iran

143 (2014). Life after Lorentz: Quantum Mechanics, Gravity, and the Crisis of Falsifiability. Physics Colloquium - University of Tehran, Tehran, Iran

144 (2013). Missing Baryons: How Much? How Hot? How Fast?. Cross-Correlating Cosmic Fields Conference, Shanghai, China

145 Gould E. (2013). A Non-Local Reality: Is there a Phase Uncertainty in Quantum Mechanics?. Quantum Foundations Seminar - Perimeter Institute for Theoretical Physics, Waterloo, Canada

146 (2013). Tales of Dark Matter on Small Scales. Colloquium - Kavli Institute for Astronomy and Astrophysics, Beijing, China

147 (2013). Dynamical Emergence of Universal Horizons During the Formation of Black Holes. LOOPS 13, Waterloo, Canada

148 (2013). Out of the White Hole: A Holographic Origin for the Big Bang. Cosmology After Planck 2013, Ann Arbor, United States of America

(2013). Where Will Einstein Fail?. CASCA 2013, Vancouver, Canada
149

(2013). Why is Particle Physics Lorentz-Invariant?. Kavli Workshop - IPMU Focus Week on Gravity and
150 Lorentz Violation, Kashiwa, Japan

(2013). Life after Lorentz: Quantum Mechanics, Gravity, and the Crisis of Falsifiability. Colloquium -
151 University of Wisconsin-Madison, Madison, United States of America

(2013). Tales of Dark Matter on Small Scales. TAMU Workshop on Dark Matter, College Station, United
152 States of America

(2012). Horava-Lifshitz Cosmology and Dark Matter Holes - 2-Lecture Series. Cape Town International
153 Cosmology School 2012, Stellenbosch, South Africa, Republic of

(2012). Hot Big Bang at a Lifshitz Point. CAP Congress 2012, Calgary, Canada
154

(2012). Reviving Gravity's Aether in Einstein's Universe. Colloquium - York University, Toronto, Canada
155

(2012). Tales of Dark Matter on Small Scales. Theoretical Physics Workshop - UWO, London, Canada
156

(2012). Why is Particle Physics Lorentz Invariant?. Experimental Search for Quantum Gravity: The Hard
157 Facts, Waterloo, Canada

(2012). Modern Cosmology. Cosmology & Gravitation Seminar - Perimeter Institute for Theoretical Physics,
158 Waterloo, Canada

Bonanno A, Brandenberger R, Magueijo J, Nemiroff R, Amelino-Camelia G. (2012). Photons and Quantum
159 Gravity in Astrophysics. Experimental Search for Quantum Gravity 2012, Waterloo, Canada

(2012). Trials and Tribulations of Aether. Conformal Nature of the Universe Conference, Waterloo, Canada
160

(2011). Where will Einstein Fail? Insights into Gravity and Dark Energy. CITA Invited Seminar - University of
161 Toronto, Toronto, Canada

(2011). Largest and Smallest Dark Matter Clusters, and What We Can Learn from Them. High Energy Theory
162 Seminar - University of Michigan, Ann Arbor, United States of America

(2011). Structure in the Phase Space and Dark Matter Astronomy. Unravelling Dark Matter Conference 2011,
163 Waterloo, Canada

(2011). Where will Einstein Fail? Insights into Gravity and Dark Energy. KICP Colloquium - University of
164 Chicago, Chicago, United States of America

(2011). Aether, Quantum Gravity and Cosmology. Philosophy of Cosmology Workshop - 11th Annual
165 Conference in the Foundations of Physics, London, Canada

(2011). Where will Einstein Fail? Insights into Gravity and Dark Energy. Vainu Bappu Memorial Gold Medal
166 Award Ceremony - Astronomical Society of India, Pune, India

(2011). Aether, Quantum Gravity and Cosmology. Cosmological Frontiers in Fundamental Physics 2011, Paris,
167 France

(2011). How can Astrophysical Compact Objects Shed Light on Dark Cosmology?. High Energy Theory
168 Seminar - University of Amsterdam, Amsterdam, Netherlands

(2011). Fine Structure of Dark Matter Phase Space. Dark Matter: Its Origin, Models, and Detection,
169 Albuquerque, United States of America

(2010). Non-Linear Structure Formation in Cosmology - 2-Part Lecture Series. Invited Lectures on Structure
170 Formation - Institute for Research in Fundamental Sciences, Tehran, Iran

(2010). First Precision Measurement in Quantum Gravity. Cosmology Seminar - Institute for Research in
171 Fundamental Sciences, Tehran, Iran

172 (2010). Stellar Black Holes and the Thermodynamic Origin of Cosmic Acceleration. Colloquium - Institut d'Astrophysique de Paris, Paris, France

173 (2010). Cosmological Constant Problem: Rethinking Quantum and Gravity. Laws of Nature: Their Nature and Knowability 2010, Waterloo, Canada

174 (2010). Cosmology and the End of Physics!. Colloquium - University of Guelph, Guelph, Canada

175 (2010). Hierarchy in the Phase Space and Dark Matter Astronomy. Astrophysics Seminar - Queen's University, Kingston, Canada

176 (2009). The End of the Cosmological Constant Problem. Astrophysics Workshop - Interactions in the Dark Sector, Leiden, Netherlands

177 (2009). Black Hole Horizons and the Origin of Cosmic Acceleration. Holographic Cosmology Workshop 2009, Waterloo, Canada

178 (2009). Gravitational Aether as a Solution to the CC Problem and more!. New Prospects for Solving the Cosmological Constant Problem 2009, Waterloo, Canada

179 (2009). Stellar Black Holes and the Origin of Cosmic Acceleration. Cosmology Seminar - KIPAC, Stanford, United States of America

180 (2009). Hierarchy in the Phase Space and Dark Matter Astronomy. CYGNUS 2009, Cambridge, United States of America

181 (2009). The End of the Cosmological Constant Problem. Cosmology Seminar - IPMU, Kashiwa, Japan

182 (2009). Stellar Black Holes and the Origin of Cosmic Acceleration. Theory Canada 5 Conference, Fredericton, Canada

183 (2009). Cuscuton and Solving the Cosmological Constant Problem. High Energy Physics Seminar - Vanderbilt University, Nashville, United States of America

184 (2009). Cuscuton and Horava-Lifshitz Gravity. Gravity at a Lifshitz Point 2009, Waterloo, Canada

185 (2009). (Towards) the End of the Cosmological Constant Problem!. Cosmology & Gravitation Seminar - Perimeter Institute for Theoretical Physics, Waterloo, Canada

186 (2008). Gravitational Aether as a Solution to the Cosmological Constant Problem. 14th International Symposium on Particles, Strings and Cosmology (PASCOS '08), Waterloo, Canada

187 (2008). Physics Beyond the Horizon. Cosmology and Gravitation Seminar - Perimeter Institute for Theoretical Physics, Waterloo, Canada

188 (2008). Physics Beyond the Horizon. Physics Colloquium - University of Pennsylvania, Philadelphia, United States of America

189 (2008). Physics Beyond the Horizon. Colloquium - Perimeter Institute for Theoretical Physics, Waterloo, Canada

190 (2008). Observational Evidence for Massive Gravity. PI CITA Joint Cosmology Day 2008, Waterloo, Canada

191 (2008). Missing Thermal Energy of the Universe. Astroparticle Workshop - Open Questions from Clusters of Galaxies, Toulouse, France

192 (2008). Primordial Non-Gaussianity, Statistics of Collapsed Objects, and the ISW Effect. Origins and Observations of Primodial Non-Gaussianity 2008, Waterloo, Canada

193 (2008). Galaxy Clusters as Cosmic Canaries. High Energy Physics Seminar - Syracuse University, Syracuse, United States of America

Broadcast Interviews

2026/07/30 - 3 Physicists on God: Aliens, String Theory, and the Limits of Science, Dr Brian Keating, YouTube; with Adam Frank.

2026/05/31 - What Came Before The Big Bang?, Apple Finch Pudding, YouTube.

2026/04/02 - The Big Bang May Not Be the Beginning: Dr. Niayesh Afshordi on Time and Nothingness, Philosophy Everyday, YouTube.

2025/10/09 - Has Science Just Proved God?, Premier Unbelievable? debate with Michel-Yves Bollore, hosted by John Nelson, YouTube.

2025/09/16 - Are We Wrong About the Big Bang?, Dr Brian Keating, YouTube.

2025/06/12 - Was There Anything Before Us?, Event Horizon, with Phil Halper, YouTube.

2025/05/26 - Mindscape 316: Niayesh Afshordi and Phil Halper on the Big Bang and Before, Sean Carroll's Mindscape, YouTube.

2025/01/19 - An Astrophysicist Breaks Down Modern Cosmology, Miles K. Donahue, YouTube.

2024/09/27 - Physicists search for Black Hole Bombs, Phil Halper's YouTube Channel, YouTube.com,

2024/10/12 https://youtu.be/bJlqWGgt2hw?si=cn9waXQiKvUHIZw3

2024/05/17 - What's inside a black hole? Fuzzballs, Echoes & The Big Bang, Phil Halper's YouTube

2024/10/12 Channel, YouTube.com, https://youtu.be/JN9jlg9AdYM?si=c4158UQDt8pRFAfn

2023/04/28 - Did the Universe Create itself ? The PTC model, Before the Big Bang 11, YouTube.com,

2024/10/12 https://youtu.be/cLSuM8rKMyk?si=xdh2appUW8wAEi-7

2023/03/17 - The Fine Tuning Argument: the critics strike back, Phil Halper's YouTube channel,

2024/10/12 YouTube.com, https://youtu.be/zNH-ZgSpBuQ?si=fahqxSe3_F4xgzDv

2022/10/09 - Echoes from the Abyss: The Mind-Bending Science Behind the Strange Sounds of

2024/10/12 Black Holes, Time before Space, YouTube.com https://www.youtube.com/watch?v=d1mcfBip_sw&ab_channel=TimebeforeSpace

2022/05/27 - Ghazal Geshnizjani & Niayesh Afshordi: Cosmologically Coupled, Conversations at

2024/10/12 Perimeter Institute, https://conversationsattheperimeter.podbean.com/

2020/10/28 - A New Model for the Covid Pandemic, Sabine Hossenfelder's channel, YouTube.com,

2024/10/12 https://youtu.be/H0oZUpMtJrs?si=TA5cEInNhZRnN5ug

2024/05/03 - Astronomers claim universe has "cosmic glitch", Phil Halper's Youtube Channel https://

2024/05/03 youtu.be/GOctev5cUCI?si=LacFQqSuTPKsAUyO, Youtube

2020/11/20 - Fighting COVID-19 with Physics, Interview with the local CTV station in Kitchener-

2021/11/20 Waterloo, CTV News, Kitchener-Waterloo

2020/11/12 - Q&A with the Experts: The problem with Herd Immunity and COVID-19, Mike Farwell Show, 570 News, Kitchener Radio

2021/11/12

2019/10/04 - Black Hole Genesis, Before the Big Bang, Episode 10, YouTube.com 2019/10/04

2018/10/17 - Varying Speed of Light Cosmology, Before the Big Bang, Episode 8, YouTube.com 2019/07/31

2017/02/10 - Big Bang and Holography, Science Friday with Ira Flatow, National Public Radio, USA 2017/02/10

2017/02/06 - Big Bang, Big Think, Radio Canada International (https://www.rcinet.ca/en/2017/02/06/the-universe-is-holographic/)

2017/02/06

2017/02/01 - Holographic Cosmology, Morning Edition, CBC Kitchener-Waterloo 2017/02/01

2017/01/01 - Enigmatic echo from Black Hole, Science@ORF.at, Austrian Broadcasting Corporation 2017/01/01

2013/10/13 - Bye Bye Big Bang?, FQXi Podcast, FQXi.org 2013/10/13

2013/03/16 - Working at PI, Fresh Air with Mary Ito, CBC Radio 2013/03/16

2012/07/04 - Discovery of Higgs Particle at CERN, 60 Minutes, BBC Persian TV 2012/07/04

2011/10/04 - Legacy of Stephen Hawking, 60 Minutes, BBC Persian TV 2011/10/04

2011/09/24 - (Claimed) Discovery of Superluminal Neutrinos, 60 Minutes, BBC Persian TV 2011/09/24

Text Interviews

2026/07/30 Science by majority, text interview by Matthew A. Bell, *Bye Bye Universe*.

2024/05/15 Newfound 'glitch' in Einstein's relativity could rewrite the rules of the universe, study suggests, LiveScience.com https://www.livescience.com/physics-mathematics/newfoundglitch-in-einsteins-relativity-could-rewrite-the-rules-of-the-universe-study-suggests

2024/05/15 A Cosmic Glitch in Gravity, Nautilus Magazine https://nautil.us/a-cosmic-glitch-ingravity-593171/?_sp=0f3668e3-5102-4de7-b03f-fc59eb4ec411.1715821798622

2024/05/01 A 'cosmic glitch' in gravity, Waterloo News https://uwaterloo.ca/news/media/cosmic-glitchgravity

2024/01/29 Over Gravity's Rainbow: An alternative framework of physics, in which light's speed can vary By Zeeya Merali, John Templeton Foundation Newsletter https://www.templeton.org/news/over-gravitys-rainbow

2023/02/17 "Dark energy from supermassive black holes? Physicists spar over radical idea", Science News https://www.science.org/content/article/dark-energy-supermassive-black-holesphysicists-spar-over-radical-idea

2021/09/20 Chiamaka Okoli was a rarity in physics. She challenged norms until her untimely death A Black woman, mother, and a devout Catholic, Okoli was a trailblazer in her field before tragedy struck, Boradview.com https://broadview.org/chiamaka-okoli/

2021/06/04 Modelling the mysteries of the universe, closer to home Professor Niayesh Afshordi used his knowledge of modelling the mysteries of the Cosmos to start modelling the mysteries of COVID-19, Waterloo News https://uwaterloo.ca/news/science/modeling-mysteriesuniverse-closer-home

2021/02/08 Echoes from black hole could force us to rethink laws of physics Gravitational echoes may confirm the late Stephen Hawking's quantum black hole hypothesis, Waterloo News, https://uwaterloo.ca/news/global-impact/echoes-black-hole-could-force-us-rethink-lawsphysics

2020/02/10 Spacetime 'Echoes' From Quantum Black Holes Could Soon Change Physics Forever, VICE Motherboard

2020/02/01 Musta aukko sateilee, TAHDET JA AVARUUS Finland

2020/01/31 Black holes shouldn't echo, but this one might. Score 1 for Stephen Hawking?, Live Science

2019/10/30 Einstein killed the aether. Now the idea is back to save relativity, New Scientist

2019/06/18 Big Bang BOMBSHELL: Are we living inside a HUGE higher dimensional black hole?, Express.co.uk, https://www.express.co.uk/news/science/1140059/black-hole-higherdimensional-big-bang-theory-universe-hologram-space-news

2019/05/24 We've Found Gravitational Waves. What Will We Learn From Them?, Discover Magazine

2019/05/15 Ripple Effect, Discover Magazine

2019/01/16 Crisis at the Heart of Physics, New Scientist

2018/06/20 Total Wipeout, New Scientist

2018/05/14 Echoes of gravitational waves could point to quantum gravity, Physics World

2017/03/29 Echoes of a Black Hole, Aeon.COM

2017/02/04 Could the universe actually be a flat hologram?, Christian Science Monitor

2017/01/31 Lost in space? A brief guide to the 'holographic principle' of the universe, Guardian

2016/12/13 Was Einstein wrong? Controversial theory claims the speed of light is not a constant, Wired Magazine

2016/11/28 Theory Challenging Einstein's View On Speed of Light Could Soon Be Tested, The Guardian

2016/09/12 LIGO black hole echoes hint at general-relativity breakdown, Nature News

2016/02/05 Much Ado around Nothing: The Cosmological non-Constant Problem, Backreaction

2016/01/11 The Sky as a Limit, Perimeter Institute News

2015/02/11 Do Black Hole Firewalls Have Observable Consequences?, Backreaction

2013/09/26 Was Our Universe Born in an Extra-Dimensional Black Hole?, PBS NOVA

2013/09/13 Did a Hyper-Black Hole Spawn the Universe?, Nature News

2013/07/16 Cosmologist Claims Universe May Not Be Expanding, Nature News

2010/07/01 Cosmic Bubble Made Cold Spot in Big Bang Afterglow, New Scientist

2009/03/16 Gravity May Venture Where Matter Fears to Tread, New Scientist

Publications

This list contains 163 scholarly records, including 157 arXiv papers. DOI links point to the published version; arXiv links point to the open manuscript.

2026

1. Adam Smith, Ilaria Andrei, Maria Mylova, Niayesh Afshordi. Cuscuton Fifth Force: Implications of Non-Dynamical Scalars for Structure Growth. Preprint (2026). arXiv:2611.01228

2. Sougato Bose, Anupam Mazumdar, Marko Toroš, Tian Zhou, Tadeusz Adach, Niayesh Afshordi, et al. White paper: 1–10 Hz matter-wave interferometer to test the spin entanglement witness for quantum gravity. Preprint (2026). arXiv:2610.07422

3. Robin Y. Wen, Lukas Tobias Hergt, Niayesh Afshordi, Douglas Scott. Is DESI Seeing Dynamical Dark Energy, or a Cosmic Glitch in Gravity? Preprint (2026). arXiv:2609.40176

4. Zachary Cheslog, Emily Finson, Amanda MacInnis, Neelima Sehgal, Niayesh Afshordi, Simran K. Nerval, Renée Hložek. Current and Future Constraints on the Primordial Power Spectrum. Preprint (2026). arXiv:2609.35964

5. Niayesh Afshordi, Kristina Giesel. A Cuscuton Representation of the Loop Quantum Cosmology Bounce. Preprint (2026). arXiv:2609.35222

6. Ruolin Liu, Niayesh Afshordi. Gauss–Bonnet running and the de Sitter saddle of quadratic gravity inflation. arXiv:2609.24912

7. Shokoufe Faraji, Niayesh Afshordi. Self-regularized entropy: What does black hole entropy predict for tests of Kerr no-hair theorem? Class. Quant. Grav. 43, 135027 (2026). doi:10.1088/1361-6382/ae7793 arXiv:2508.20421

8. Guandi Zhao, Alex Krolewski, Niayesh Afshordi. Thermal Sunyaev-Zel'dovich cross-correlations with unWISE galaxies: disentangling radio contamination, dust properties, and electron pressure. arXiv:2606.28099

9. Ruijing Tang, Shokoufe Faraji, Niayesh Afshordi. Exploring Born-Infeld f(T) teleparallel gravity through accretion disk dynamics. Phys. Rev. D 113, 123021 (2026). doi:10.1103/6q8h-k57n arXiv:2502.03775

10. Niayesh Afshordi, Phil Halper, Matteo Rini, Michael Schirber. Big Mysteries Survey: Physicists' Views on Cosmology, Black Holes, Quantum Mechanics, and Quantum Gravity. arXiv:2605.11058

11. Ruolin Liu, Jerome Quintin, Niayesh Afshordi. Ultraviolet Completion of the Big Bang in Quadratic Gravity. Phys. Rev. Lett. 136, 111501 (2026). doi:10.1103/6gtx-j455 arXiv:2510.18733

12. Emanuele Berti, Vitor Cardoso, Gregorio Carullo, Jahed Abedi, Niayesh Afshordi, Simone Albanesi, Vishal Baibhav, Swetha Bhagwat, et al.. Black hole spectroscopy: from theory to experiment. Class. Quant. Grav. 43, 123001 (2026). doi:10.1088/1361-6382/ae59e2 arXiv:2505.23895

2025

13. Jiayue Yang, Niayesh Afshordi, Mahdi Torabian, Seyed Akbar Jafari, G. Baskaran. Smart Holes: Analogue black holes with the right temperature and entropy. JHEP 07, 226 (2025). doi:10.1007/jhep07(2025)226 arXiv:2412.08517

14. Alice Chen, Niayesh Afshordi. Where are all the dark galaxies? Predicting galaxy/halo locations from their bright neighbors. arXiv:2507.01814

15. Martin Radler, Niayesh Afshordi, Reza Taleei, Katia Parodi, Ramin Abolfath, Julie Lascaud. A semiclassical model of the immediate temperature distribution surrounding the track of heavy ions with therapeutic energies. Physics in Medicine & Biology. doi:10.1088/1361-6560/add83b

16. Alice Y. Chen, Phil Halper, Niayesh Afshordi. Copenhagen Survey on Black Holes and Fundamental Physics. arXiv:2503.15776

17. Batia Friedman-Shaw, Alex Krolewski, Matteo Foglieni, Niayesh Afshordi. Doppler bias: impact of peculiar velocities on color selection and the large scale structure of galaxy surveys. JCAP 03, 059 (2025). doi:10.1088/1475-7516/2025/03/059 arXiv:2410.04705

[18] Ruolin Liu, Jerome Quintin, Niayesh Afshordi. Conformal Cores of Quantum Black Holes in Quadratic Gravity. Phys. Rev. D 111, 044031 (2025). doi:10.1103/physrevd.111.044031 arXiv:2411.19311

[19] Conner Dailey, Erik Schnetter, Niayesh Afshordi. Formulating the complete initial boundary value problem in numerical relativity to model black hole echoes. Class. Quant. Grav. 42, 025002 (2025). doi:10.1088/1361-6382/ad9701 arXiv:2409.17970

[20] Lakshay Goel, Everett A. Patterson, Maria Rosa Preciado-Rivas, Mahdi Torabian, Robert B. Mann, Niayesh Afshordi. Accelerated detector in a superposed spacetime. Phys. Rev. D 111, 025015 (2025). doi:10.1103/physrevd.111.025015 arXiv:2409.06818

[21] Robin Y. Wen, Lukas T. Hergt, Niayesh Afshordi, Douglas Scott. A glitch in gravity: cosmic Lorentz-violation from fiery Big Bang to glacial heat death. Astrophys. Space Sci. Proc. 61, 435 (2025). doi:10.1007/978-3-031-90186-7_35 arXiv:2412.09568

[22] LISA Consortium Waveform Working Group, Niayesh Afshordi, Sarp Akcay, Pau Amaro Seoane, Andrea Antonelli, Josu C. Aurrekoetxea, Leor Barack, Enrico Barausse, et al.. Waveform Modelling for the Laser Interferometer Space Antenna. Living Rev. Rel. 28, 9 (2025). doi:10.1007/s41114-025-00056-1 arXiv:2311.01300

[23] Niayesh Afshordi, Stefano Liberati. Quantum black holes: from regularization to information paradoxes. Gen. Rel. Grav. 57, 25 (2025). doi:10.1007/s10714-025-03356-7

2024

[24] Sedighe Sajadian, Niayesh Afshordi. Simulating Self-Lensing and Eclipsing Signals due to Detached Compact Objects in the TESS Light Curves. The Astronomical Journal. doi:10.3847/1538-3881/ad7fdd arXiv:2409.12441

[25] Niayesh Afshordi, Abhay Ashtekar, Enrico Barausse, Emanuele Berti, Richard Brito, Luca Buoninfante, Raul Carballo-Rubio, Vitor Cardoso, et al.. Black Holes Inside and Out 2024: visions for the future of black hole physics. arXiv:2410.14414

[26] Yu Li, Ruolin Liu, Conner Dailey, Niayesh Afshordi. Detecting cosmological scalar fields using orbital networks of quantum sensors. Phys. Rev. D 110, 043028 (2024). doi:10.1103/physrevd.110.043028 arXiv:2311.17873

[27] Ramin Abolfath, Niayesh Afshordi, Sohrab Rahvar, Adri van Duin, Martin Radler, Reza Taleei, Katia Parodi, Julie Lascaud, Radhe Mohan. A molecular dynamics simulation of the abrupt changes in the thermodynamic properties of water after formation of nano-bubbles / nano-cavities induced by passage of charged particles. Eur. Phys. J. D (2024) 78: 141. doi:10.1140/epjd/s10053-024-00928-1 arXiv:2403.05880

[28] M. Mylova, N. Afshordi. Effective Cuscuton Theory. JHEP 04, 144 (2024). doi:10.1007/jhep04(2024)144 arXiv:2312.06066

[29] Robin Y. Wen, Lukas T. Hergt, Niayesh Afshordi, Douglas Scott. A cosmic glitch in gravity. JCAP 03, 045 (2024). doi:10.1088/1475-7516/2024/03/045 arXiv:2311.03028

[30] Ramit Dey, Luis Felipe Longo Micchi, Suvodip Mukherjee, Niayesh Afshordi. Spectrogram correlated stacking: A novel time-frequency domain analysis of the Stochastic Gravitational Wave Background. Phys. Rev. D 109, 023029 (2024). doi:10.1103/physrevd.109.023029 arXiv:2305.03090

2023

[31] Conner Dailey, Niayesh Afshordi, Erik Schnetter. Reflecting boundary conditions in numerical relativity as a model for black hole echoes. Class. Quant. Grav. 40, 195007 (2023). doi:10.1088/1361-6382/acde2f arXiv:2301.05778

[32] Jahed Abedi, Luis Felipe Longo Micchi, Niayesh Afshordi. GW190521: Search for Echoes due to Stimulated Hawking Radiation from Black Holes. Phys. Rev. D 108, 044047 (2023). doi:10.1103/physrevd.108.044047 arXiv:2201.00047

[33] Alice Chen, Niayesh Afshordi. On the Outskirts of Dark Matter Haloes. Phys. Rev. D 107, 103526 (2023). doi:10.1103/physrevd.107.103526 arXiv:2210.11499

[34] Di Wu, Pengyuan Gao, Jing Ren, Niayesh Afshordi. Model-independent search for the quasinormal modes of gravitational wave echoes. Phys. Rev. D 108, 124006 (2023). doi:10.1103/physrevd.108.124006 arXiv:2308.01017

[35] Naritaka Oshita, Niayesh Afshordi. From Entropy to Echoes: Counting the quasi-normal modes and the quantum limit of silence. Phys. Lett. B 841, 137901 (2023). doi:10.1016/j.physletb.2023.137901 arXiv:2302.08964

[36] Johanna N. Borissova, Aaron Held, Niayesh Afshordi. Scale-invariance at the core of quantum black holes. Class. Quant. Grav. 40, 075011 (2023). doi:10.1088/1361-6382/acbc60 arXiv:2203.02559

2022

[37] Niayesh Afshordi, Joao Magueijo. Lower bound on the cosmological constant from the classicality of the Early Universe. Phys. Rev. D 106, 123518 (2022). doi:10.1103/physrevd.106.123518 arXiv:2209.07914

[38] Matthew P. G. Robbins, Niayesh Afshordi, Alan O. Jamison, Robert B. Mann. Detection of Gravitational Waves using Parametric Resonance in Bose-Einstein Condensates. Class. Quant. Grav. 39, 175009 (2022). doi:10.1088/1361-6382/ac7b05 arXiv:2101.03691

[39] Ramin Abolfath, Alexander Baikalov, Stefan Bartzsch, Niayesh Afshordi, Radhe Mohan. The effect of non-ionizing excitations on the diffusion of ion species and inter-track correlations in FLASH ultra-high dose rate radiotherapy. Physics in Medicine & Biology 67, 105005 (2022). doi:10.1088/1361-6560/ac69a6 arXiv:2201.07887

[40] K. G. Arun, Enis Belgacem, Robert Benkel, Laura Bernard, Emanuele Berti, Gianfranco Bertone, Marc Besancon, Diego Blas, et al. (including Niayesh Afshordi). New Horizons for Fundamental Physics with LISA. Living Rev. Rel. 25, 4 (2022). doi:10.1007/s41114-022-00036-9 arXiv:2205.01597

[41] Tianyi Yang, Michael J. Hudson, Niayesh Afshordi. A Universal Profile for Stacked Filaments from Cold Dark Matter Simulations. Mon. Not. Roy. Astron. Soc. 516, 6041 (2022). doi:10.1093/mnras/stac2564 arXiv:2203.16170

[42] Maria Mylova, Marianthi Moschou, Niayesh Afshordi, Joao Magueijo. Non-Gaussian Signatures of a Thermal Big Bang. JCAP 07, 005 (2022). doi:10.1088/1475-7516/2022/07/005 arXiv:2112.08179

[43] Krishan Saraswat, Niayesh Afshordi. Spacing Statistics of Energy Spectra: Random Matrices, Black Hole Thermalization, and Echoes. JHEP 04, 017 (2022). doi:10.1007/jhep04(2022)017 arXiv:2110.03188

2021

[44] Randy S. Conklin, Niayesh Afshordi. Boltzmann Meets Lorentz: A Surrogate Model for Black Hole Echoes. arXiv:2201.00027

[45] Luis Felipe Longo Micchi, Niayesh Afshordi, Cecilia Chirenti. How loud are echoes from exotic compact objects? Phys. Rev. D 103, 044028 (2021). doi:10.1103/physrevd.103.044028 arXiv:2010.14578

[46] Wan Zhen Chua, Niayesh Afshordi. Electromagnetic Albedo of Quantum Black Holes. JHEP 07, 006 (2021). doi:10.1007/jhep07(2021)006 arXiv:2103.05790

[47] Naritaka Oshita, Niayesh Afshordi, Shinji Mukohyama. Lifshitz scaling, ringing black holes, and superradiance. JCAP 05, 005 (2021). doi:10.1088/1475-7516/2021/05/005 arXiv:2102.01741

[48] Peter Adshead, Niayesh Afshordi, Emanuela Dimastrogiovanni, Matteo Fasiello, Eugene A. Lim, Gianmassimo Tasinato. Multimessenger Cosmology: correlating CMB and SGWB measurements. Phys. Rev. D 103, 023532 (2021). doi:10.1103/physrevd.103.023532 arXiv:2004.06619

[49] Krishan Saraswat, Niayesh Afshordi. Extracting Hawking Radiation Near the Horizon of AdS Black Holes. JHEP 02, 077 (2021). doi:10.1007/jhep02(2021)077 arXiv:2003.12676

2020

[50] Naritaka Oshita, Daichi Tsuna, Niayesh Afshordi. Quantum Black Hole Seismology I: Echoes, Ergospheres, and Spectra. Phys. Rev. D 102, 024045 (2020). doi:10.1103/physrevd.102.024045 arXiv:2001.11642

51 Niayesh Afshordi, Benjamin Holder, Mohammad Bahrami, Daniel Lichtblau. Diverse local epidemics reveal the distinct effects of population density, demographics, climate, depletion of susceptibles, and intervention in the first wave of COVID-19 in the United States. doi:10.1101/2020.06.30.20143636 arXiv:2007.00159

52 Alice Y. Chen, Niayesh Afshordi. Amending the halo model to satisfy cosmological conservation laws. Phys. Rev. D 101, 103522 (2020). doi:10.1103/physrevd.101.103522 arXiv:1912.04872

53 Jahed Abedi, Niayesh Afshordi, Naritaka Oshita, Qingwen Wang. Quantum Black Holes in the Sky. Universe 6, 43 (2020). doi:10.3390/universe6030043 arXiv:2001.09553

54 Jahed Abedi, Niayesh Afshordi. Echoes from the Abyss: A Status Update. arXiv:2001.00821

55 Ramit Dey, Niayesh Afshordi. Echoes in the Kerr/CFT correspondence. Phys. Rev. D 102, 126006 (2020). doi:10.1103/physrevd.102.126006 arXiv:2009.09027

56 Naritaka Oshita, Daichi Tsuna, Niayesh Afshordi. Quantum Black Hole Seismology II: Applications to Astrophysical Black Holes. Phys. Rev. D 102, 024046 (2020). doi:10.1103/physrevd.102.024046 arXiv:2004.06276

57 Tianyi Yang, Michael J. Hudson, Niayesh Afshordi. How dark are filaments in the cosmic web? Mon. Not. Roy. Astron. Soc. 498, 3158 (2020). doi:10.1093/mnras/staa2547 arXiv:2001.10943

58 Enrico Barausse, Emanuele Berti, Thomas Hertog, Scott A. Hughes, Philippe Jetzer, Paolo Pani, Thomas P. Sotiriou, Nicola Tamanini, et al. (including Niayesh Afshordi). Prospects for Fundamental Physics with LISA. Gen. Rel. Grav. 52, 81 (2020). doi:10.1007/s10714-020-02691-1 arXiv:2001.09793

59 Ramit Dey, Sumanta Chakraborty, Niayesh Afshordi. Echoes from the braneworld black holes. Phys. Rev. D 101, 104014 (2020). doi:10.1103/physrevd.101.104014 arXiv:2001.01301

60 Tianyi. Yang, Supranta Sarma Boruah, Niayesh. Afshordi. Gravitational Potential from small-scale clustering in action space: Application to Gaia DR2. Mon. Not. Roy. Astron. Soc. 493, 3061 (2020). doi:10.1093/mnras/staa441 arXiv:1908.02336

61 Krishan Saraswat, Niayesh Afshordi. Quantum Nature of Black Holes: Fast Scrambling versus Echoes. JHEP 04, 136 (2020). doi:10.1007/jhep04(2020)136 arXiv:1906.02653

62 Naritaka Oshita, Qingwen Wang, Niayesh Afshordi. On Reflectivity of Quantum Black Hole Horizons. JCAP 04, 016 (2020). doi:10.1088/1475-7516/2020/04/016 arXiv:1905.00464

63 Qingwen Wang, Naritaka Oshita, Niayesh Afshordi. Echoes from Quantum Black Holes. Phys. Rev. D 101, 024031 (2020). doi:10.1103/physrevd.101.024031 arXiv:1905.00446

2019

64 Niayesh Afshordi. On the origin of the LIGO "mystery" noise and the high energy particle physics desert. arXiv:1911.09384

65 Elizabeth Gould, Niayesh Afshordi. Does History Repeat Itself? Periodic Time Cosmology. JCAP 09, 058 (2019). doi:10.1088/1475-7516/2019/09/058 arXiv:1903.09694

66 Matthew P. G. Robbins, Niayesh Afshordi, Robert B. Mann. Bose-Einstein Condensates as Gravitational Wave Detectors. JCAP 07, 032 (2019). doi:10.1088/1475-7516/2019/07/032 arXiv:1811.04468

2018

67 Nosiphiwo Zwane, Niayesh Afshordi, Rafael D. Sorkin. Cosmological Tests of Everpresent $Lambda$. Class. Quant. Grav. 35, 194002 (2018). doi:10.1088/1361-6382/aadc36 arXiv:1703.06265

68 Chiamaka Okoli, James E. Taylor, Niayesh Afshordi. Searching for dark matter annihilation from individual halos: uncertainties, scatter and signal-to-noise ratios. JCAP 08, 019 (2018). doi:10.1088/1475-7516/2018/08/019 arXiv:1711.05271

69 Naritaka Oshita, Niayesh Afshordi. Probing microstructure of black hole spacetimes with gravitational wave echoes. Phys. Rev. D 99, 044002 (2019). doi:10.1103/physrevd.99.044002 arXiv:1807.10287

70 Niayesh Afshordi, Matthew C. Johnson. Cosmological Zero Modes. Phys. Rev. D 98, 023541 (2018). doi:10.1103/physrevd.98.023541 arXiv:1708.04694

[71] Mohammad H. Zhoolideh Haghighi, Niayesh Afshordi, Habib. G. Khosroshahi. Cooling+Heating Flows in Galaxy clusters: Turbulent heating, spectral modelling, and cooling efficiency. Astrophys. J. 884, 47 (2019). doi:10.3847/1538-4357/ab379c arXiv:1806.08822

[72] Qingwen Wang, Niayesh Afshordi. Black Hole Echology: The Observer's Manual. Phys. Rev. D 97, 124044 (2018). doi:10.1103/physrevd.97.124044 arXiv:1803.02845

[73] Mansour Karami, Niayesh Afshordi, Jesus Zavala. Forward modelling of quasar light curves and the cosmological matter power spectrum on milliparsec scales. ArXiv e-prints. arXiv:1805.06984

[74] Aditya Dhumuntarao, Jose Tomas Galvez Ghersi, Niayesh Afshordi. Instantaneous Temperatures a la Hadamard: Towards a generalized Stefan-Boltzmann law for curved spacetime. ArXiv e-prints. arXiv:1804.05382

[75] Jahed Abedi, Niayesh Afshordi. Echoes from the Abyss: A highly spinning black hole remnant for the binary neutron star merger GW170817. JCAP 11, 010 (2019). doi:10.1088/1475-7516/2019/11/010 arXiv:1803.10454

[76] Jahed Abedi, Hannah Dykaar, Niayesh Afshordi. Comment on: "Low significance of evidence for black hole echoes in gravitational wave data". ArXiv e-prints. arXiv:1803.08565

2017

[77] Nima Khosravi, Shant Baghram, Niayesh Afshordi, Natacha Altamirano. H_0 tension as a hint for a transition in gravitational theory. Phys. Rev. D 99, 103526 (2019). doi:10.1103/physrevd.99.103526 arXiv:1710.09366

[78] Jahed Abedi, Hannah Dykaar, Niayesh Afshordi. Echoes from the Abyss: Tentative evidence for Planck-scale structure at black hole horizons. Phys. Rev. D 96, 082004 (2017). doi:10.1103/physrevd.96.082004 arXiv:1612.00266

[79] Yasaman K. Yazdi, Niayesh Afshordi. Accretion in Radiative Equipartition (AiRE) Disks. Astrophys. J. 843, 22 (2017). doi:10.3847/1538-4357/aa73d4 arXiv:1609.06716

[80] Niayesh Afshordi, Elizabeth Gould, Kostas Skenderis. Constraining holographic cosmology using Planck data. Phys. Rev. D 95, 123505 (2017). doi:10.1103/physrevd.95.123505 arXiv:1703.05385

[81] Chiamaka Okoli, Morag I. Scrimgeour, Niayesh Afshordi, Michael J. Hudson. Dynamical friction in the primordial neutrino sea. Mon. Not. Roy. Astron. Soc. 468, 2164 (2017). doi:10.1093/mnras/stx560 arXiv:1611.04589

[82] Niayesh Afshordi, Hyungjin Kim, Elliot Nelson. Pulsar Timing Constraints on Physics Beyond the Standard Model. ArXiv e-prints. arXiv:1703.05331

[83] Natacha Altamirano, Elizabeth Gould, Niayesh Afshordi, Robert B. Mann. Cosmological Perturbations in the 5D Holographic Big Bang Model. Fundam. Theor. Phys. 219, 205 (2025). doi:10.1007/978-3-031-76066-2_10 arXiv:1703.00954

[84] Mehdi Saravani, Niayesh Afshordi. Off-shell Dark Matter: A Cosmological relic of Quantum Gravity. Phys. Rev. D 95, 043514 (2017). doi:10.1103/physrevd.95.043514 arXiv:1604.02448

[85] Jahed Abedi, Hannah Dykaar, Niayesh Afshordi. Echoes from the Abyss: The Holiday Edition! ArXiv e-prints. arXiv:1701.03485

[86] Niayesh Afshordi, Claudio Coriano, Luigi Delle Rose, Elizabeth Gould, Kostas Skenderis. From Planck data to Planck era: Observational tests of Holographic Cosmology. Phys. Rev. Lett. 118, 041301 (2017). doi:10.1103/physrevlett.118.041301 arXiv:1607.04878

2016

[87] Niayesh Afshordi, Yasaman K. Yazdi. "Firewall" Phenomenology with Astrophysical Neutrinos. Class. Quant. Grav. 33, 235017 (2016). doi:10.1088/0264-9381/33/23/235017 arXiv:1502.01023

[88] Morgan H. Lynch, Niayesh Afshordi. Temperatures of renormalizable quantum field theories in curved spacetime. Class. Quant. Grav. 35, 225008 (2018). doi:10.1088/1361-6382/aae792 arXiv:1611.06619

[89] Niayesh Afshordi, Joao Magueijo. The critical geometry of a thermal big bang. Phys. Rev. D 94, 101301 (2016). doi:10.1103/physrevd.94.101301 arXiv:1603.03312

90 Michael Meiers, Mehdi Saravani, Niayesh Afshordi. Cosmic Censorship in Lorentz Violating Theories of Gravity. Phys. Rev. D 93, 104008 (2016). doi:10.1103/physrevd.93.104008 arXiv:1511.08969

91 Niayesh Afshordi, Elliot Nelson. Cosmological bounds on TeV-scale physics and beyond. Phys. Rev. D 93, 083505 (2016). doi:10.1103/physrevd.93.083505 arXiv:1504.00012

92 Chiamaka Okoli, Niayesh Afshordi. Concentration, Ellipsoidal Collapse, and the Densest Dark Matter haloes. Mon. Not. Roy. Astron. Soc. 456, 3068 (2016). doi:10.1093/mnras/stv2905 arXiv:1510.03868

93 Jesus Zavala, Niayesh Afshordi. Universal clustering of dark matter in phase space. Mon. Not. Roy. Astron. Soc. 457, 986 (2016). doi:10.1093/mnras/stw048 arXiv:1508.02713

2015

94 Elizabeth S. Gould, Niayesh Afshordi. A Non-Local Reality: Is there a Phase Uncertainty in Quantum Mechanics? Found. Phys. 45, 1620 (2015). doi:10.1007/s10701-015-9948-8 arXiv:1407.4083

95 Niayesh Afshordi. Why is High Energy Physics Lorentz Invariant? ArXiv e-prints. arXiv:1511.07879

96 N. Afshordi, R.B. Mann, R. Pourhasan. A holographic big bang? Int. J. Mod. Phys. D 24, 1544029 (2015). doi:10.1142/s0218271815440290

97 Farbod Kamiab, Avery E. Broderick, Niayesh Afshordi. The mass and radii of strongly magnetized neutron stars. ArXiv e-prints. arXiv:1503.03898

2014

98 Niayesh Afshordi, Dejan Stojkovic. Emergent Spacetime in Stochastically Evolving Dimensions. Phys. Lett. B 739, 117 (2014). doi:10.1016/j.physletb.2014.10.048 arXiv:1405.3297

99 Niayesh Afshordi, Michele Fontanini, Daniel C. Guariento. Horndeski meets McVittie: A scalar field theory for accretion onto cosmological black holes. Phys. Rev. D 90, 084012 (2014). doi:10.1103/physrevd.90.084012 arXiv:1408.5538

100 Takeshi Kobayashi, Niayesh Afshordi. Schwinger Effect in 4D de Sitter Space and Constraints on Magnetogenesis in the Early Universe. JHEP 10, 166 (2014). doi:10.1007/jhep10(2014)166 arXiv:1408.4141

101 Abhineet Agarwal, Niayesh Afshordi. Thermal Tachyacoustic Cosmology. Phys. Rev. D 90, 043528 (2014). doi:10.1103/physrevd.90.043528 arXiv:1406.0575

102 Ali Narimani, Niayesh Afshordi, Douglas Scott. How does pressure gravitate? Cosmological constant problem confronts observational cosmology. JCAP 08, 049 (2014). doi:10.1088/1475-7516/2014/08/049 arXiv:1406.0479

103 Jesus Zavala, Niayesh Afshordi. Clustering in the Phase Space of Dark Matter Haloes. II. Stable Clustering and Dark Matter Annihilation. Mon. Not. Roy. Astron. Soc. 441, 1329 (2014). doi:10.1093/mnras/stu506 arXiv:1311.3296

104 Jesus Zavala, Niayesh Afshordi. Clustering in the Phase Space of Dark Matter Haloes. I. Results from the Aquarius simulations. Mon. Not. Roy. Astron. Soc. 441, 1317 (2014). doi:10.1093/mnras/stu678 arXiv:1308.1098

105 Elcio Abdalla, Niayesh Afshordi, Michele Fontanini, Daniel C. Guariento, Eleftherios Papantonopoulos. Cosmological black holes from self-gravitating fields. Phys. Rev. D 89, 104018 (2014). doi:10.1103/physrevd.89.104018 arXiv:1312.3682

106 Mehdi Saravani, Niayesh Afshordi, Robert B. Mann. Dynamical Emergence of Universal Horizons during the formation of Black Holes. Phys. Rev. D 89, 084029 (2014). doi:10.1103/physrevd.89.084029 arXiv:1310.4143

107 Razieh Pourhasan, Niayesh Afshordi, Robert B. Mann. Out of the White Hole: A Holographic Origin for the Big Bang. JCAP 04, 005 (2014). doi:10.1088/1475-7516/2014/04/005 arXiv:1309.1487

108 Sohrab Rahvar, Shant Baghram, Niayesh Afshordi. Transient Weak-Lensing by Cosmological Dark Matter Microhaloes. Phys. Rev. D 89, 063001 (2014). doi:10.1103/physrevd.89.063001 arXiv:1310.5412

2013

109 Siavash Aslanbeigi, Guilhem Lavaux, Amir Hajian, Niayesh Afshordi. An Optimal and Model-Independent Measurement of the Intracluster Pressure Profile I: Methodology and First Applications. Mon. Not. Roy. Astron. Soc. 435, 1788 (2013). doi:10.1093/mnras/stt1416 arXiv:1304.6457

110 Guilhem Lavaux, Niayesh Afshordi, Michael J. Hudson. First measurement of the bulk flow of nearby galaxies using the cosmic microwave background. Mon. Not. Roy. Astron. Soc. 430, 1617 (2013). doi:10.1093/mnras/sts698 arXiv:1207.1721

111 Niayesh Afshordi. An Optimal & Model-Independent Measurement of the Pressure Profile of Hot Gas in Clusters of Galaxies Using the SZ Effect. SnowCLUSTER 2013, Physics of Galaxy Clusters.

2012

112 Mehdi Saravani, Niayesh Afshordi, Robert B. Mann. Empty Black Holes, Firewalls, and the Origin of Bekenstein-Hawking Entropy. Int. J. Mod. Phys. D 23, 1443007 (2015). doi:10.1142/s021827181443007x arXiv:1212.4176

113 Jesus Zavala, Michael L. Balogh, Niayesh Afshordi, Stephen Ro. Removal and mixing of the coronal gas from satellites in galaxy groups: cooling the intragoup gas. Mon. Not. Roy. Astron. Soc. 426, 3464 (2012). doi:10.1111/j.1365-2966.2012.21980.x arXiv:1208.4849

114 Niayesh Afshordi, Michel Buck, Fay Dowker, David Rideout, Rafael D. Sorkin, Yasaman K. Yazdi. A Ground State for the Causal Diamond in 2 Dimensions. JHEP 10, 088 (2012). doi:10.1007/jhep10(2012)088 arXiv:1207.7101

115 Maxim Pospelov, Niayesh Afshordi. Lithium Diffusion in the Post-Recombination Universe and Spatial Variation of [Li/H]. ArXiv e-prints. arXiv:1208.0793

116 Niayesh Afshordi, Siavash Aslanbeigi, Rafael D. Sorkin. A Distinguished Vacuum State for a Quantum Field in a Curved Spacetime: Formalism, Features, and Cosmology. JHEP 08, 137 (2012). doi:10.1007/jhep08(2012)137 arXiv:1205.1296

117 Niayesh Afshordi. Where will Einstein fail? Lessons for gravity and cosmology. Bull. Astron. Soc. India 40, 1 (2012). arXiv:1203.3827

2011

118 R. Pourhasan, N. Afshordi, R. B. Mann, A. C. Davis. Chameleon Gravity, Electrostatics, and Kinematics in the Outer Galaxy. JCAP 12, 005 (2011). doi:10.1088/1475-7516/2011/12/005 arXiv:1109.0538

119 Siavash Aslanbeigi, Georg Robbers, Brendan Z. Foster, Kazunori Kohri, Niayesh Afshordi. Phenomenology of Gravitational Aether as a solution to the Old Cosmological Constant Problem. Phys. Rev. D 84, 103522 (2011). doi:10.1103/physrevd.84.103522 arXiv:1106.3955

120 Farbod Kamiab, Niayesh Afshordi. Neutron Stars and the Cosmological Constant Problem. Phys. Rev. D 84, 063011 (2011). doi:10.1103/physrevd.84.063011 arXiv:1104.5704

121 Shant Baghram, Niayesh Afshordi, Kathryn M. Zurek. Prospects for Detecting Dark Matter Halo Substructure with Pulsar Timing. Phys. Rev. D 84, 043511 (2011). doi:10.1103/physrevd.84.043511 arXiv:1101.5487

122 Niayesh Afshordi, Anze Slosar, Yi Wang. A Theory of a Spot. JCAP 01, 019 (2011). doi:10.1088/1475-7516/2011/01/019 arXiv:1006.5021

2010

123 C. Prescod-Weinstein, N. Afshordi. Using Dark Matter Haloes to Learn about Cosmic Acceleration: A New Proposal for a Universal Mass Function. ArXiv e-prints. arXiv:1010.5501

124 Bruce A. Bassett, Niayesh Afshordi. Non-Gaussian Posteriors arising from Marginal Detections. ArXiv e-prints. arXiv:1005.1664

125 Niayesh Afshordi, Roya Mohayaee, Edmund Bertschinger. Hierarchy in the Phase Space and Dark Matter Astronomy. Phys. Rev. D 81, 101301 (2010). doi:10.1103/physrevd.81.101301 arXiv:0911.0414

126 Niayesh Afshordi. Reviving Gravity's Aether in Einstein's Universe. ArXiv e-prints. arXiv:1004.2901

Niayesh Afshordi. Dark Energy, Black Hole Entropy, and the First Precision Measurement in Quantum Gravity. ArXiv e-prints. arXiv:1003.4811 [127]

S. Ahlen, N. Afshordi, J. B. R. Battat, J. Billard, N. Bozorgnia, S. Burgos, T. Caldwell, J. M. Carmona, et al.. The case for a directional dark matter detector and the status of current experimental efforts. Int. J. Mod. Phys. A 25, 1 (2010). doi:10.1142/s0217751x10048172 arXiv:0911.0323 [128]

2009

Niayesh Afshordi. Cuscuton and low energy limit of Horava-Lifshitz gravity. Phys. Rev. D 80, 081502 (2009). doi:10.1103/physrevd.80.081502 arXiv:0907.5201 [129]

Chanda Prescod-Weinstein, Niayesh Afshordi, Michael L. Balogh. Stellar Black Holes and the Origin of Cosmic Acceleration. Phys. Rev. D 80, 043513 (2009). doi:10.1103/physrevd.80.043513 arXiv:0905.3551 [130]

Niayesh Afshordi, Ghazal Geshnizjani, Justin Khoury. Observational Evidence for Cosmological-Scale Extra Dimensions. JCAP 08, 030 (2009). doi:10.1088/1475-7516/2009/08/030 arXiv:0812.2244 [131]

Niayesh Afshordi, Roya Mohayaee, Edmund Bertschinger. Hierarchical Phase Space Structure of Dark Matter Haloes: Tidal debris, Caustics, and Dark Matter annihilation. Phys. Rev. D 79, 083526 (2009). doi:10.1103/physrevd.79.083526 arXiv:0811.1582 [132]

E. Komatsu, N. Afshordi, N. Bartolo, D. Baumann, J. R. Bond, E. I. Buchbinder, C. T. Byrnes, X. Chen, et al.. Non-Gaussianity as a Probe of the Physics of the Primordial Universe and the Astrophysics of the Low Redshift Universe. astro2010: The Astronomy and Astrophysics Decadal Survey. arXiv:0902.4759 [133]

2008

Marilena LoVerde, Niayesh Afshordi. Extended Limber Approximation. Phys. Rev. D 78, 123506 (2008). doi:10.1103/physrevd.78.123506 arXiv:0809.5112 [134]

Niayesh Afshordi, Andrew J. Tolley. Primordial non-gaussianity, statistics of collapsed objects, and the Integrated Sachs-Wolfe effect. Phys. Rev. D 78, 123507 (2008). doi:10.1103/physrevd.78.123507 arXiv:0806.1046 [135]

Niayesh Afshordi. Fundamental Plane of Sunyaev-Zeldovich clusters. Astrophys. J. 686, 201 (2008). doi:10.1086/591307 arXiv:0704.2416 [136]

Niayesh Afshordi. Gravitational Aether and the thermodynamic solution to the cosmological constant problem. ArXiv e-prints. arXiv:0807.2639 [137]

Georg Robbers, Niayesh Afshordi, Michael Doran. Does Planck mass run on the cosmological horizon scale? Phys. Rev. Lett. 100, 111101 (2008). doi:10.1103/physrevlett.100.111101 arXiv:0708.3235 [138]

2007

Niayesh Afshordi, Daniel J. H. Chung, Michael Doran, Ghazal Geshnizjani. Cuscuton Cosmology: Dark Energy meets Modified Gravity. Phys. Rev. D 75, 123509 (2007). doi:10.1103/physrevd.75.123509 arXiv:astro-ph/0702002 [139]

Niayesh Afshordi, Yen-Ting Lin, Daisuke Nagai, Alastair J. R. Sanderson. Missing Thermal Energy of the Intracluster Medium. Mon. Not. Roy. Astron. Soc. 378, 293 (2007). doi:10.1111/j.1365-2966.2007.11776.x arXiv:astro-ph/0612700 [140]

Niayesh Afshordi, Daniel J. H. Chung, Ghazal Geshnizjani. Cuscuton: A Causal Field Theory with an Infinite Speed of Sound. Phys. Rev. D 75, 083513 (2007). doi:10.1103/physrevd.75.083513 arXiv:hep-th/0609150 [141]

Niayesh Afshordi. How well can (renormalized) perturbation theory predict dark matter clustering properties? Phys. Rev. D 75, 021302 (2007). doi:10.1103/physrevd.75.021302 arXiv:astro-ph/0610336 [142]

2006

Niayesh Afshordi. Intracluster Medium through three years of WMAP. New Astron. Rev. 50, 905 (2006). doi:10.1016/j.newar.2006.09.022 arXiv:astro-ph/0608503 [143]

Niayesh Afshordi. CMB Constraints on the Intracluster Medium. External Correlations of the CMB and Cosmology. [144]

145 Banibrata Mukhopadhyay, Niayesh Afshordi, Ramesh Narayan. Growth of Hydrodynamic Perturbations in Accretion Disks: Possible Route to Non-Magnetic Turbulence. Advances in Space Research. doi:10.1016/j.asr.2005.09.048 arXiv:astro-ph/0507046

2005

146 Niayesh Afshordi, Matias Zaldarriaga, Kazunori Kohri. On the stability of Dark Energy with Mass-Varying Neutrinos. Phys. Rev. D 72, 065024 (2005). doi:10.1103/physrevd.72.065024 arXiv:astro-ph/0506663

147 Niayesh Afshordi, Banibrata Mukhopadhyay, Ramesh Narayan. Bypass to Turbulence in Hydrodynamic Accretion: Lagrangian Analysis of Energy Growth. Astrophys. J. 629, 373 (2005). doi:10.1086/431418 arXiv:astro-ph/0412194

148 Banibrata Mukhopadhyay, Niayesh Afshordi, Ramesh Narayan. Bypass to Turbulence in Hydrodynamic Accretion Disks: An Eigenvalue Approach. Astrophys. J. 629, 383 (2005). doi:10.1086/431419 arXiv:astro-ph/0412193

149 Niayesh Afshordi, Yen-Ting Lin, Alastair J. R. Sanderson. WMAP constraints on the Intra-Cluster Medium. Astrophys. J. 629, 1 (2005). doi:10.1086/431274 arXiv:astro-ph/0408560

150 Ghazal Geshnizjani, Daniel J. H. Chung, Niayesh Afshordi. Do Large-Scale Inhomogeneities Explain Away Dark Energy? Phys. Rev. D 72, 023517 (2005). doi:10.1103/physrevd.72.023517 arXiv:astro-ph/0503553

151 Christopher M. Hirata, Abraham Loeb, Niayesh Afshordi. CMB B-mode polarization from Thomson scattering in the local universe. Phys. Rev. D 71, 063531 (2005). doi:10.1103/physrevd.71.063531 arXiv:astro-ph/0501167

152 Ghazal Geshnizjani, Niayesh Afshordi. Coarse-Grained Back Reaction in Single Scalar Field Driven Inflation. JCAP 01, 011 (2005). doi:10.1088/1475-7516/2005/01/011 arXiv:gr-qc/0405117

2004

153 Niayesh Afshordi. Integrated Sachs-Wolfe effect in Cross-Correlation: The Observer's Manual. Phys. Rev. D 70, 083536 (2004). doi:10.1103/physrevd.70.083536 arXiv:astro-ph/0401166

154 Banibrata Mukhopadhyay, Niayesh Afshordi, Ramesh Narayan. Hydrodynamic Turbulence in Accretion Disks. eConf C041213, 1609 (2004). arXiv:astro-ph/0501468

2003

155 Niayesh Afshordi, Yeong-Shang Loh, Michael A. Strauss. Cross-Correlation of the Cosmic Microwave Background with the 2MASS Galaxy Survey: Signatures of Dark Energy, Hot Gas, and Point Sources. Phys. Rev. D 69, 083524 (2004). doi:10.1103/physrevd.69.083524 arXiv:astro-ph/0308260

156 Niayesh Afshordi. Cross-Correlating the CMB with Large-Scale Structure. KITP: Astrophysics Seminars.

157 N. Afshordi, P. McDonald, D. N. Spergel. Primordial Black Holes as Dark Matter: The Power Spectrum and Evaporation of Early Structures. Astrophys. J. Lett. 594, L71 (2003). doi:10.1086/378763 arXiv:astro-ph/0302035

158 R. Scranton, A. J. Connolly, R. C. Nichol, A. Stebbins, I. Szapudi, D. J. Eisenstein, N. Afshordi, T. Budavari, et al.. Physical Evidence for Dark Energy. ArXiv e-prints. arXiv:astro-ph/0307335

159 Niayesh Afshordi, Bohdan Paczynski. Geometrically Thin Disk Accreting Into a Black Hole. Astrophys. J. 592, 354 (2003). doi:10.1086/375559 arXiv:astro-ph/0202409

2002

160 Niayesh Afshordi. Cyclotron Emission and Thermalization of the CMB Spectrum. ArXiv e-prints. arXiv:astro-ph/0202082

161 Niayesh Afshordi, Renyue Cen. Mass-Temperature Relation of Galaxy Clusters: A Theoretical Study. Astrophys. J. 564, 669 (2002). doi:10.1086/324282 arXiv:astro-ph/0105020

2001

162 Niayesh Afshordi, Robert Brandenberger. Super-Hubble Nonlinear Perturbations During Inflation. Phys. Rev. D 63, 123505 (2001). doi:10.1103/physrevd.63.123505 arXiv:gr-qc/0011075

1999

163 E. Abdalla, N. Afshordi, K. Khodjasteh, R. Mohayaee. A Dynamical Approach to a Self-similar Universe. Astron. Astrophys. 345, 22 (1999). arXiv:astro-ph/9803187

Books and public scholarship

***Battle of the Big Bang: The New Tales of Our Cosmic Origins*, with Phil Halper** — University of Chicago Press, 2025

"The Black Hole That Birthed the Big Bang," with Robert B. Mann and Razieh Pourhasan — *Scientific American*, 2014

Research software

P2SAD: Particle Phase Space Average Density, with Jesus Zavala — Public research code, 2014